Classical Field Theory

Lecture Notes

Fields, symmetries, variational
principles, and physical laws.

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Introduction

Purpose and Character of These Notes

Classical field theory is often presented in an extremely compressed form. A few symbols are introduced, an action is written down, a variation is performed in one line, and a field equation appears. To an experienced reader, such a presentation may be efficient. To a student meeting the subject for the first time, it can make the theory look like a collection of formulas that arrive without a clear reason, operate by hidden rules, and disappear before their meaning has been examined.

These notes are designed to avoid that style of exposition. Their purpose is not merely to collect standard formulas from analytical mechanics, special relativity, electrodynamics, and gauge theory. Their purpose is to show how the main constructions of classical field theory are built, why they are introduced, what assumptions they use, how the calculations are carried out, and how the final expressions are interpreted physically and mathematically.

A formula will therefore not be treated as the end of an explanation. Whenever possible, it will be presented as one stage in a sequence:

1. identify the physical or mathematical problem;
2. determine what objects must be introduced;
3. state the assumptions and conventions;
4. derive the relevant relation;
5. test it on a simple model;
6. interpret the result;
7. examine limiting cases and possible failures.

The central principle of the course is that a student should not only recognize an equation, but should know what to do with it. For example, it is not enough to state the Euler–Lagrange field equations. A student should see how the variation of a field is defined, how derivatives of the variation arise, where integration by parts is used, why a boundary term disappears, how indices are managed, and how the resulting equation is applied to an explicit Lagrangian density. Similarly, it is not enough to write down Noether's theorem. The theorem must be used to construct an actual current, verify its divergence, and identify the associated conserved charge.

The same standard will be applied to electrodynamics. The electromagnetic field tensor will not be introduced as an isolated matrix to memorize. It will be built from the four-potential, its components will be related explicitly to the electric and magnetic fields, its transformation properties will be examined, and its invariants will be calculated. Maxwell's equations will then be derived from an action, rather than merely restated in covariant notation.

These notes are meant to serve two related purposes. During the lecture, they provide a structured path through the material and a reliable source of notation, derivations, and examples. After the lecture, they should function as a readable reconstruction of the argument. A student who did not follow one step on the board should be able to return to the notes and see how that step fits into the complete calculation.

The exposition will therefore contain several recurring elements.

Motivation before formalism

A new object will be introduced because it solves a problem or expresses a principle. The field concept will be motivated by locality and continuous degrees of freedom. The action will be motivated as a compact way to encode dynamics. Four-dimensional notation will be motivated by Lorentz symmetry. Gauge freedom will be motivated by the non-uniqueness of the potentials that describe the same physical electromagnetic field.

Definitions connected to examples

Definitions will be followed by concrete cases. Scalar fields will be represented by temperature, density, or a real dynamical field. Vector fields will be connected with velocity fields and electric fields. Tensor fields will be introduced through objects that are actually used later, rather than through formal classification alone.

Complete derivations

Important formulas will be derived in sufficient detail that the logical structure is visible. This does not mean that every elementary algebraic step must be expanded indefinitely. It means that no essential conceptual step should be hidden behind phrases such as "it is easy to show" or "after some manipulation." When a sign, boundary condition, index contraction, or change of variables matters, it will be displayed.

Simple proofs and checks

Not every statement in physics requires a theorem-proof format, but important claims should be justified. We will prove elementary identities when they clarify the structure of the theory, verify conservation laws directly, check gauge invariance explicitly, and compare covariant formulas with familiar three-dimensional expressions. Dimensional analysis, symmetry checks, and limiting cases will be used as routine diagnostic tools.

Worked calculations

The notes will include explicit calculations rather than only general rules. A free scalar field will be solved by a plane-wave ansatz. A Noether current will be computed for a specified transformation. The electromagnetic field tensor will be written component by component. The Green function for the Poisson equation will be applied to a point source. The radiation field of an oscillating dipole will be developed far enough to calculate the angular distribution and total emitted power.

Physical interpretation

After a derivation, we will ask what the result means. What quantity is conserved? What propagates? Which variables are measurable? Which variables contain redundancy? What changes under a transformation, and what remains invariant? A correct calculation without interpretation is incomplete, because field theory is not only a formal language but also a way of organizing physical information.

Connections between chapters

The course is designed as a chain of constructions. The action principle for particles becomes the action principle for fields. Translation symmetry becomes energy-momentum conservation. The scalar field provides a simple model before the electromagnetic field is introduced. The electromagnetic potential becomes the first example of a gauge connection. The field strength becomes the first example of curvature. These connections will be stated explicitly so that the course does not appear as twelve unrelated topics.

The level of the notes is therefore deliberately intermediate between a concise reference text and a fully expanded textbook. They will remain mathematically precise, but they will also explain why the mathematics is being used. They will contain derivations and examples, but they will avoid turning every section into a long catalogue of exercises. The objective is a coherent set of lecture notes that can be read, checked, and used.

How Each Topic Will Be Developed

To keep the exposition consistent, most substantial topics will be developed according to a common pattern. The details will vary, but the reader should repeatedly encounter the same sequence of questions: what problem are we solving, what structures are needed, how is the result derived, and how can it be used?

1. The motivating problem

A section should begin with a recognizable task or conceptual difficulty. We may want to describe a continuous system, derive a differential equation from an action, express a conservation law locally, solve a field equation with a source, or understand why two different potentials represent the same electromagnetic field. This opening question determines the purpose of the formalism that follows.

The motivation should be specific enough to guide the calculation. Instead of saying only that fields are important, we will ask how a system with one degree of freedom at every point in space can be described. Instead of saying only that symmetries imply conservation laws, we will ask how an infinitesimal transformation changes the action and how that change produces a divergence equation.

2. The objects, assumptions, and conventions

Before calculating, the relevant objects must be defined. Their domains, codomains, indices, units, transformation rules, and regularity assumptions should be clear. In relativistic calculations, the metric signature and index conventions must be fixed. In variational calculations, the allowed variations and boundary conditions must be stated. In Green-function methods, the differential operator and boundary conditions must be identified.

This stage is essential because many apparent contradictions in field theory are actually differences of convention. A sign in the metric changes signs in scalar products. A definition of the field-strength tensor changes the placement of signs in its components. A choice of units changes the constants appearing in Maxwell's equations. The notes will not hide these choices.

3. The derivation

The main relation will then be derived step by step. The derivation should show every operation that carries conceptual information. If a derivative is moved from a variation to another factor, the integration by parts will be written. If dummy indices are renamed, the reason will be visible. If antisymmetry causes a term to vanish, the antisymmetry will be used explicitly. If a boundary term is discarded, the corresponding boundary condition will be stated.

A representative example is the field-theoretic action

$$S[\phi] = \int_\Omega \mathcal{L}(\phi, \partial_\mu \phi, x)\, \mathrm{d}^4x.$$

The notes will not jump directly from this expression to the Euler–Lagrange field equation. They will define a varied field $\phi + \varepsilon\eta$, differentiate with respect to ε, separate the variation of the field from the variation of its derivative, integrate by parts, analyze the boundary contribution, and only then infer the differential equation.

4. A proof, verification, or structural check

Once a formula has been derived, it will be checked. The nature of the check depends on the result. A proposed current may be shown to have zero divergence on solutions of the field equation. A covariant Maxwell equation may be expanded into its three-dimensional components. A gauge transformation may be applied directly to verify that the field tensor remains unchanged. A stress-energy tensor may be tested by computing its energy-density component.

These checks serve two purposes. They confirm the algebra, and they reveal what the abstract expression contains. A tensor formula becomes much easier to understand after one calculates its individual components and recovers a familiar physical law.

5. A fully worked example

General formulas become useful only when they are applied. Each major construction should therefore be followed by at least one complete example, and usually by several examples of increasing complexity.

For a scalar field, we may begin with the free massless field, derive the wave equation, substitute a plane wave, and obtain the dispersion relation. We may then add a mass term and compare the result. For Noether's theorem, we may first treat spacetime translations and then a global phase transformation of a complex scalar field. For Green functions, we may begin with a point source and then pass to a distributed source by integration.

A worked example should include the setup, the calculation, and the interpretation. It should not end immediately after the final algebraic expression. The reader should be told what the answer says about propagation, conservation, symmetry, or physical measurement.

6. Limiting cases and alternative forms

An important result should be examined in limits where its meaning is already known. A relativistic formula may be reduced to a nonrelativistic one. A massive scalar field may be compared with the massless case. A retarded solution may be compared with a static solution. A tensor equation may be rewritten in vector notation.

Alternative forms will also be compared when this improves understanding. The electromagnetic field may be described using $\mathbf{E}$ and $\mathbf{B}$, the four-potential A_μ, the antisymmetric tensor $F_{\mu\nu}$, or a differential two-form. These descriptions are not competing theories; they emphasize different structural features of the same theory.

7. Common mistakes and diagnostic questions

Where a calculation is especially sensitive to signs, indices, or assumptions, the notes will identify typical errors. Examples include confusing raised and lowered components, varying a derivative incorrectly, discarding a boundary term without justification, treating gauge-dependent quantities as observables, or using a static Green function for a time-dependent problem.

The reader should learn not only a successful calculation, but also how to detect an unsuccessful one. Useful questions include:

- Are the free indices the same on both sides of the equation?
- Do the dimensions agree?
- Is the expression invariant under the symmetry it is supposed to respect?
- Does the result reduce correctly in a simple limit?
- Have all boundary conditions been used consistently?
- Is the conserved quantity really independent of time?

8. A concise conclusion

A section should close by identifying what has been established and what will be used next. This final paragraph is important in a cumulative subject. The student should know which result is a definition, which is a derived equation, which is a conservation law, and which is a method of solution.

The notes will therefore distinguish clearly between several kinds of statements:

- definitions that introduce notation or structure;
- assumptions that restrict the model;
- identities that hold algebraically;
- field equations that hold for physical solutions;
- conservation laws that follow from symmetries;
- examples that illustrate a method;
- approximations that hold only in a specified regime.

This distinction prevents formulas from appearing as an undifferentiated list. It allows the reader to understand not only what an equation says, but why it is valid and when it may be used.

Course Structure and Chapter Roadmap

The course is organized into twelve principal chapters, corresponding approximately to twelve teaching weeks. The chapters are not independent units. Each one introduces tools that are used later, and the order is chosen so that the formalism grows from familiar mechanics toward relativistic fields, electrodynamics, propagation, radiation, and geometric field theories.

The first three chapters establish the field concept and the variational method. Chapters 4–6 introduce the relativistic language, the scalar-field prototype, and the relation between symmetry and conservation. Chapters 7–9 develop electromagnetism as a complete classical field theory, including gauge freedom and energy-momentum. Chapters 10–11 explain how fields generated by sources are solved and how radiation emerges. Chapter 12 gives a geometric outlook toward gauge theory and gravity.

Chapter 1: Fields, Locality, and Continuous Degrees of Freedom

The opening chapter establishes what a field is and why fields are required. We begin with familiar examples: temperature as a scalar field, the velocity of a fluid as a vector field, the displacement of a string as a time-dependent field, and the electric field as a physical quantity defined throughout space. These examples make clear that a field assigns data to every point of a domain and therefore carries infinitely many degrees of freedom.

The idea of locality is then introduced. A local field theory describes changes at a point in terms of the field and its derivatives near that point. This contrasts with a direct action-at-a-distance description. The distinction will be illustrated through sources, responses, and propagation. We will ask what it means for a source to affect a field locally and how information can move through a continuous medium.

A central worked construction will be the passage from a discrete chain of coupled oscillators to a continuous string. The discrete labels will be replaced by a spatial coordinate, finite differences will become derivatives, and the sum over masses will become an integral. This example will show explicitly how a field theory can emerge as the continuum limit of a many-particle system.

By the end of the chapter, the student should be able to identify scalar, vector, and tensor fields, state their domains and values, distinguish discrete from continuous degrees of freedom, and explain why local differential equations are natural equations of motion for fields.

Chapter 2: Action Principles: From Particles to Fields

The second chapter develops the variational principle in the setting of ordinary mechanics. This is necessary because the field-theoretic action is a direct extension of the mechanical action. We introduce a functional as an object whose input is an entire trajectory rather than a single number. The action is then defined as an integral of the Lagrangian along a path.

The variation of a trajectory will be constructed explicitly. We will compare a physical trajectory with nearby trajectories that share fixed endpoints, differentiate the action with respect to the variation parameter, and use integration by parts. The boundary term will not simply be discarded; the endpoint conditions that make it vanish will be stated and interpreted.

The Euler–Lagrange equations will be derived in full and applied to several concrete systems: a free particle, a particle in a potential, the harmonic oscillator, and a simple constrained coordinate. These examples will show how to move from a Lagrangian to an equation of motion and then verify that a proposed solution satisfies the resulting equation.

This chapter also establishes a working habit that will recur throughout the course: whenever an equation is derived from an action, the allowed variations, the boundary data, and the variables held fixed must be specified.

Chapter 3: Lagrangian Field Theory and Field Equations

The third chapter performs the transition from trajectories to fields. The Lagrangian becomes a Lagrangian density, and the action becomes an integral over spacetime. A variation now changes the field value at every point, and derivatives of the field produce derivatives of the variation.

The Euler–Lagrange field equations will be derived carefully for one field and then generalized to several coupled fields. The calculation will display the spacetime integration by parts and the boundary term on the boundary of the integration region. This provides the basic method used in nearly every later derivation.

Several field equations will be obtained from actions. The vibrating string will produce the wave equation. A static energy functional will lead to Laplace or Poisson equations. A field with a potential $V(\phi)$ will provide a first example of a nonlinear field equation. We will compare the roles of kinetic, gradient, source, and potential terms in a Lagrangian density.

The chapter will close by identifying canonical field momenta and the Hamiltonian density at an introductory level. The main goal, however, is practical: given a Lagrangian density, the student should be able to vary it, manage the boundary term, and derive the corresponding differential equation.

Chapter 4: Spacetime, Lorentz Symmetry, and Covariant Notation

Classical relativistic field theory requires a language in which space and time are combined consistently. This chapter introduces Minkowski spacetime, events, the spacetime interval, proper time, and Lorentz transformations. The metric convention will be fixed and used throughout the remaining chapters.

Four-vectors, covectors, and tensors will be introduced through calculations rather than definitions alone. Students will practice raising and lowering indices, contracting tensors, distinguishing free and dummy indices, and checking whether an expression is a Lorentz scalar. Four-position, four-velocity, four-momentum, and the four-gradient will provide the main examples.

The d'Alembert operator will be constructed from the metric and the four-gradient. Familiar equations such as the wave equation will then be rewritten in covariant form. We will compare the compact tensor notation with the expanded time-and-space components so that the notation does not conceal the underlying differential equation.

Particular attention will be given to sign conventions. The notes will show how the chosen metric signature affects norms, derivatives, and Lagrangian densities. By the end of the chapter, the student should be able to read and manipulate the index notation used in the remainder of the course.

Chapter 5: The Real Scalar Field

The real scalar field is the simplest complete relativistic field theory and will serve as the principal training model. We construct the free scalar-field Lagrangian from Lorentz scalars and derive the massless and massive field equations. The massive equation is the classical Klein–Gordon equation, treated here as a classical partial differential equation rather than as a quantum equation.

Plane-wave solutions will be substituted explicitly into the field equation. This calculation will produce the dispersion relation and clarify the meaning of frequency, wave vector, and the mass parameter. The massless and massive cases will be compared, including phase and group behavior where appropriate.

The chapter will then examine energy and momentum for scalar fields. We will calculate the energy density of simple configurations and ask under what conditions the energy is bounded below. Interacting theories with a potential $V(\phi)$ will be introduced, and equilibrium configurations will be identified as extrema of the potential.

Linearization around an equilibrium will show how a nonlinear theory produces an approximate linear wave equation for small perturbations. This provides a useful method that reappears in many areas of physics: find a background solution, perturb it, keep the leading terms, and analyze stability.

Chapter 6: Symmetries, Currents, and Noether's Theorem

This chapter develops one of the central organizing principles of field theory: continuous symmetries produce local conservation laws. We begin with infinitesimal transformations of coordinates and fields, making clear what changes actively and what changes because the coordinates have been relabeled.

Noether's theorem will be derived for fields with enough detail to identify every term in the current. The derivation will distinguish variations that vanish at the boundary from transformations that represent physical symmetries. The resulting divergence equation will be related to a conserved charge by integration over space.

Several explicit applications will follow. Time translation will be connected with energy, spatial translation with momentum, and rotations with angular momentum. A complex scalar field will provide an example of an internal global phase symmetry and its associated current. For each case, the current will be calculated and its conservation checked using the field equation.

The chapter will also prepare the transition to gauge theory by distinguishing a global symmetry from a local redundancy. The student should leave with a practical algorithm: specify the infinitesimal transformation, compute the variation of the Lagrangian density, identify the Noether current, and verify its divergence.

Chapter 7: Electromagnetic Potentials and Gauge Freedom

Electromagnetism is introduced through potentials because the potentials are the natural variables of the action principle. We begin with the scalar and vector potentials and derive the electric and magnetic fields from them. The relations will be checked against the homogeneous

Maxwell equations.

Gauge transformations will then be performed explicitly. The potentials change, but the electric and magnetic fields remain unchanged. This calculation provides the first concrete example of a description containing more variables than there are physical degrees of freedom. The distinction between a change of description and a change of physical state will be emphasized.

The scalar and vector potentials will be combined into the four-potential A_μ. From it we construct the antisymmetric electromagnetic field tensor $F_{\mu\nu}$. The matrix of components will be written out, and the electric and magnetic components will be identified with careful attention to signs and index positions.

The two standard Lorentz invariants of the electromagnetic field will be calculated and interpreted. Finally, the homogeneous Maxwell equations will be written in covariant form and related to the fact that the field tensor is constructed from a potential.

Chapter 8: Maxwell Theory from an Action

This chapter builds Maxwell theory as a variational field theory. The electromagnetic Lagrangian density is formed from the field tensor, and a coupling term to an external four-current is added. We then vary the action with respect to the four-potential.

The calculation will be shown in full: the variation of the field tensor, the use of antisymmetry, the integration by parts, the treatment of the boundary term, and the extraction of the inhomogeneous Maxwell equations. The covariant result will be expanded into Gauss's law and the Ampere–Maxwell law.

Gauge invariance of the source coupling will be analyzed. We will show that the action changes by a boundary term only when the current is conserved. This makes the relation between gauge invariance and charge conservation explicit rather than merely asserted.

The Lorenz gauge condition will be introduced and used to reduce the equations for the potentials to wave equations. The action for a relativistic charged particle interacting with the electromagnetic potential will then be varied to obtain the Lorentz-force law. This completes a unified action-based description of both the field and its interaction with matter.

Chapter 9: Energy, Momentum, and Stress in Field Theory

Field equations describe evolution, but they do not by themselves show how energy and momentum are distributed and transported. This chapter develops the local energy-momentum description of scalar and electromagnetic fields.

We begin with canonical field momentum and Hamiltonian density. Translation symmetry is then used to construct the canonical energy-momentum tensor. The distinction between canonical and symmetric tensors will be discussed, including why the symmetric form is especially natural in relativistic theories and in coupling to gravity.

For the scalar field, the energy density, momentum density, and flux will be computed. For the electromagnetic field, the components of the energy-momentum tensor will be identified with electromagnetic energy density, the Poynting vector, momentum density, and the Maxwell stress tensor.

Several explicit calculations will show how these quantities are used. We will calculate energy flow in a plane wave, force from the stress tensor in a simple geometry, and radiation pressure. Local conservation equations will be integrated over a finite region to obtain balance laws with flux through the boundary.

Chapter 10: Green Functions, Sources, and Field Propagation

The tenth chapter turns from deriving field equations to solving them. Green functions provide a systematic method for linear differential equations with sources. We introduce the idea through an operator equation and define a Green function as the response to a point source.

The static case will be developed first. The Green function of the Poisson equation in three-dimensional space will be used to recover the Coulomb potential. The solution for a distributed source will then be obtained by superposition. This calculation will connect the abstract Green-function definition with a familiar physical result.

Time-dependent equations require additional choices. We will construct retarded and advanced Green functions for wave equations and explain the boundary conditions encoded by each choice. The retarded solution will be associated with causal propagation from past sources to later fields.

The method will then be applied to electromagnetic potentials. Retarded potentials will be derived as integrals over the source evaluated at retarded time. The chapter will emphasize that a differential equation alone does not determine a unique physical solution; boundary and causal conditions are part of the problem.

Chapter 11: Radiation from Time-Dependent Sources

Radiation is the regime in which a time-dependent source creates a field that carries energy to arbitrarily large distances. We begin by separating near-field terms from radiation-field terms according to their dependence on distance. This makes clear why a $1/r$ field can transport a finite amount of energy through a large sphere.

A multipole expansion will organize the radiation produced by localized sources. The electric dipole will be studied in detail. Starting from an oscillating dipole moment, we will calculate the far-zone fields, the Poynting vector, the angular distribution of emitted power, and the total radiated power.

The Lienard–Wiechert potentials will then be introduced for a moving point charge. Their dependence on retarded time will be analyzed, and the velocity and acceleration contributions to the field will be distinguished. The acceleration field will be identified as the radiative part.

The Larmor formula will be derived or motivated from the radiation field, with its assumptions stated clearly. Energy balance and radiation reaction will be discussed at an introductory level, including the conceptual difficulties that arise when a point charge interacts with its own field.

Chapter 12: Gauge Fields and Gravity: The Geometric Outlook

The final chapter places the preceding material in a broader geometric framework. Electromagnetism will be reinterpreted as a $U(1)$ gauge theory. The four-potential will be viewed as a connection, the field tensor as curvature, and the gauge-covariant derivative as the derivative appropriate for charged fields.

This language will be introduced through the electromagnetic case before any non-Abelian generalization is attempted. The non-Abelian field strength will then be presented as a natural extension in which the gauge potentials themselves interact because the symmetry group is noncommutative.

Gravity will be introduced as a theory in which the metric is a dynamical field. Connections, curvature, geodesics, and the Einstein–Hilbert action will be discussed at a structural level. The variation of the gravitational action will be outlined sufficiently to show why the Einstein tensor appears and why stress-energy acts as a source.

The weak-field form $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ will connect general relativity with linear field theory and the Newtonian potential. The chapter will conclude by comparing electromagnetism and gravity: their dynamical variables, symmetries, sources, field strengths, conservation laws, and geometric interpretations. The purpose is not to complete a course in Yang–Mills theory or general relativity, but to show where the methods developed in the previous eleven chapters lead.

How to Read and Extend These Notes

These notes are intended to complement a lecture presented at the board. The lecture can emphasize the main physical argument, develop a representative derivation, and respond to questions. The written notes can preserve the complete structure: definitions, conventions, intermediate steps, calculations, checks, diagrams, and conclusions. Neither form should merely duplicate the other. Together they should allow the student to see both the main line of reasoning and the details needed for independent study.

A productive way to read a chapter is to separate three levels of understanding.

1. **Conceptual level:** What physical or mathematical problem is being addressed? What new object or principle is introduced?

2. **Derivational level:** How does the main equation follow from the assumptions? Which steps use algebra, calculus, symmetry, or boundary conditions?

3. **Operational level:** Given a specific field, source, or transformation, can the method be applied to obtain a concrete result?

The student should not try to memorize every displayed equation at the first reading. It is more useful to identify the role of each equation. Is it a definition, an identity, an equation of motion, a conservation law, a gauge choice, or a solution under specified boundary conditions? Once that role is understood, the formula is easier to reconstruct and use.

Calculations should be read actively. Before following a derivation, the reader should note the variables, free indices, dimensions, and expected form of the answer. After the derivation, the result should be checked in a simple case. For instance, a covariant equation can be expanded into components, a conservation law can be integrated over space, and a relativistic expression can be examined in a low-velocity limit.

The notes will be developed incrementally. Each chapter is divided into section files, and the substantive content will be written one section at a time. This is not only a repository-management choice. It supports mathematical and pedagogical quality. A single section can be checked for logical continuity, notation, sign conventions, examples, and links to earlier material before the next section is added.

The intended development cycle for future work is:

1. select one section;

2. identify its precise learning objective;

3. write the motivation and definitions;

4. develop the derivation or central argument;

5. add one or more explicit examples;

6. verify notation, equations, and limiting cases;

7. compile and inspect the PDF;

8. continue to the next section only after the current one is coherent.

This section-by-section method prevents a chapter from becoming a long sequence of loosely connected formulas. It also makes it possible to revise the course structure without rewriting unrelated material. If a topic later requires more detail, its section can be expanded or divided. If a topic proves secondary, it can be shortened without disturbing the rest of the chapter.

The twelve chapters describe the intended logical route, but the final notes should remain responsive to the actual needs of students. Some ideas may require additional preliminary explanations. Some calculations may need a second example. A diagram may communicate a geometric relation more effectively than another paragraph. The structure should therefore be stable enough to guide the course, but flexible enough to improve as the material is taught and reviewed.

A successful use of these notes should lead to more than recognition. After studying a topic, the student should be able to explain the problem in words, reproduce the main derivation with stated assumptions, carry out a representative calculation, interpret the result, and recognize the most common mistakes. That standard will guide the construction of every subsequent section.

Chapter 1

Fields, Locality, and Continuous Degrees of Freedom

1.1 What Is a Field?

A field is a way of assigning physical or mathematical information to every point of a region. The word “field” is familiar from phrases such as temperature field, velocity field, electric field, or gravitational field, but the common structure behind these examples is easy to miss. In each case, the question is not only “what is the value of a quantity?” but rather

> What value does the quantity have *here*, and how does that value compare with the value at nearby points?

This spatial dependence is the first essential feature of a field. If the field can also change with time, then we ask the same question at every point and at every instant.

From a single quantity to a distributed quantity

Consider the temperature of a small object that is kept almost uniform. In a simple model, one number may be sufficient:

$$T = 293\,\mathrm{K}.$$

This number describes the whole object only because we have assumed that the temperature is approximately the same everywhere. If one end of the object is heated, that assumption is no longer reasonable. We now need a temperature value for each position. For a thin rod, we may write

$$T = T(x, t),$$

where x labels position along the rod and t denotes time. The symbol T still represents temperature, but it no longer denotes one number. It denotes a rule that returns a number when a position and a time are supplied.

For example, suppose that at one instant the temperature along a one-metre rod is approximated by

$$T(x) = 293\,\mathrm{K} + 10\,\mathrm{K\,m^{-1}}\,x, \qquad 0 \le x \le 1\,\mathrm{m}.$$

At the left end,

$$T(0) = 293\,\mathrm{K},$$

while at $x = 0.40\,\mathrm{m}$,

$$T(0.40\,\mathrm{m}) = 293\,\mathrm{K} + 10\,\mathrm{K\,m^{-1}} \cdot 0.40\,\mathrm{m} = 297\,\mathrm{K}.$$

The calculation is elementary, but it shows the operational meaning of the field: the formula assigns a temperature to every allowed position. A single expression contains an entire spatial distribution.

A mathematical description

Abstractly, a field may be written as a map

$$\Phi : D \longrightarrow V.$$

Here:

- D is the domain on which the field is defined;
- V is the set or space of possible field values;
- $\Phi(p)$ is the field value at the point $p \in D$.

The domain may be a line, a surface, a region of three-dimensional space, or spacetime itself. The value space V may consist of real numbers, vectors, matrices, tensors, or more complicated objects. The distinction between these possibilities will be developed in the next section.

The notation should always reveal what information the field requires. For a static field in ordinary space we may write

$$\Phi = \Phi(\mathbf{x}),$$

where $\mathbf{x} = (x, y, z)$. For a time-dependent field we write

$$\Phi = \Phi(\mathbf{x}, t).$$

Later, in relativistic notation, space and time will be combined into a spacetime point x^μ, and the same field may be written as $\Phi(x^\mu)$.

A field configuration

At a fixed time $t = t_0$, the function

$$\mathbf{x} \longmapsto \Phi(\mathbf{x}, t_0)$$

is called a field configuration, or a snapshot of the field. It tells us the value of the field everywhere in space at that instant.

This is the field-theoretic analogue of specifying the position of a particle at one instant. The important difference is that a particle position is described by finitely many numbers, whereas a field configuration requires one value at every point of the spatial domain.

For a vibrating string, for example, the displacement field may be denoted by

$$u = u(x, t).$$

At a fixed time $t = t_0$, the graph of $u(x, t_0)$ is the actual shape of the string at that instant. Knowing only $u(0.5\,\mathrm{m}, t_0)$ tells us the displacement of one point, not the configuration of the whole string. To specify the complete state of the string, we need the displacement for every x, and generally also its time derivative $\partial u/\partial t$.

This observation is the origin of the statement that a field has continuously many degrees of freedom. The idea will be made precise later in this chapter by comparing a field with a chain of many coupled oscillators.

A field is more than a collection of values

It is useful to distinguish a field from an arbitrary list of unrelated numbers. Physical fields are usually assumed to vary in a sufficiently regular way. Nearby points tend to have related field values, and derivatives such as

$$\frac{\partial \Phi}{\partial x}, \qquad \frac{\partial \Phi}{\partial t}, \qquad \nabla \Phi$$

carry physical information.

For the temperature field, a spatial derivative measures how rapidly temperature changes from one place to another. For the displacement of a string, a spatial derivative measures the local slope, while a time derivative measures the local velocity. These derivatives allow a theory to connect the behaviour at one point with the behaviour in its neighbourhood.

A field theory therefore contains more than the statement that a field exists. It also contains equations that determine how the field may vary. A typical field equation relates the value of the field and its derivatives. For example, the one-dimensional wave equation has the form

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}.$$

We will derive this equation later rather than accept it as a formula to memorize. For the moment, its structure is what matters: the acceleration of one small part of the string is related to the curvature of the string near that point.

Background fields and dynamical fields

Not every field appearing in a calculation is treated in the same way. A useful distinction is the following.

A *background field* is prescribed as part of the environment. We may study the motion of a charged particle in a given electric field $\mathbf{E}(\mathbf{x}, t)$ without asking how that electric field was produced.

A *dynamical field* is an unknown quantity whose evolution is determined by a field equation. In electrodynamics, the electromagnetic field itself becomes dynamical when Maxwell's equations are solved together with specified sources and boundary data.

The same mathematical object can play either role depending on the problem. A gravitational field may be treated as a fixed background when studying a falling particle, or as a dynamical field when solving the equations of general relativity. It is therefore important to ask:

> Is this field given, or is it one of the unknowns of the theory?

Fields and measurement

A field is not identified only by a symbol. Its physical meaning depends on how its values are connected with observations. A temperature field may be sampled by thermometers placed at different positions. A fluid velocity field may be inferred from the motion of small tracer particles. An electric field may be defined operationally through the force on a sufficiently small test charge.

This does not mean that every useful field variable is directly observable. Potentials, for example, may contain a redundancy: different potentials can describe the same electric and magnetic fields. Such cases require care, because a mathematical variable may be indispensable for formulating the theory even when not every part of it represents an independent observable quantity.

For each field introduced in this course, we will therefore ask four questions:

1. On what domain is the field defined?

2. What kind of object is assigned to each point?

3. Is the field prescribed or dynamical?

4. How are its values connected with measurable effects?

What has been established

A field is a rule that assigns an appropriate value to every point of a domain. A time-dependent field assigns such values throughout space and throughout time. A configuration is the complete spatial distribution at one instant, while the full field history describes how configurations change.

The essential conceptual step is the replacement of a finite list of variables by a continuously distributed quantity. The next section examines what kind of value may be attached to each point and why scalar, vector, and tensor fields must be distinguished.

1.2 Scalar, Vector, and Tensor Fields

The previous section described a field as a rule that assigns a value to every point of a domain. We now ask a second question:

> What kind of value is attached to each point?

The answer matters because different kinds of values respond differently when coordinates are changed. A temperature does not acquire a direction when the coordinate axes are rotated. A velocity does have a direction, although its numerical components depend on the chosen axes. A stress field contains still more information: it describes how forces act on surfaces with different orientations.

The terms *scalar*, *vector*, and *tensor* identify these different transformation behaviours.

Scalar fields

A scalar field assigns one number to every point. In three-dimensional space, a scalar field may be written as

$$\phi : D \subseteq \mathbb{R}^3 \longrightarrow \mathbb{R}, \qquad \mathbf{x} \longmapsto \phi(\mathbf{x}).$$

Typical examples include temperature, mass density, pressure, electric potential, and the real scalar field studied later in this course.

The word *scalar* does not merely mean that we have written one numerical component. It means that the physical value is unchanged when we describe the same point using another coordinate system. If a room has temperature $295\,\mathrm{K}$ at a particular event, rotating the coordinate axes does not change that temperature.

As a simple example, consider the field on a flat plate

$$T(x, y) = 300\,\mathrm{K} - 2\,\mathrm{K\,m^{-1}}\,x + 3\,\mathrm{K\,m^{-1}}\,y.$$

At the point $(x, y) = (1\,\mathrm{m}, 2\,\mathrm{m})$,

$$T(1\,\mathrm{m}, 2\,\mathrm{m}) = 300\,\mathrm{K} - 2\,\mathrm{K} + 6\,\mathrm{K} = 304\,\mathrm{K}.$$

Another observer may use rotated coordinates (x', y'), but the temperature assigned to the same physical point is still $304\,\mathrm{K}$. The formula written in terms of x' and y' may look different, yet the scalar value at the point is the same.

A useful distinction is therefore:

- the *coordinate expression* of a field may change;
- the physical scalar assigned to a point does not depend on the coordinate labels.

Vector fields

A vector field assigns a vector to every point:

$$\mathbf{v} : D \subseteq \mathbb{R}^3 \longrightarrow \mathbb{R}^3.$$

Examples include the velocity field of a fluid, the electric field, the magnetic field, and the displacement field of an elastic medium.

In a chosen Cartesian basis, a vector field is written in components as

$$\mathbf{v}(\mathbf{x}) = v^1(\mathbf{x})\,\mathbf{e}_1 + v^2(\mathbf{x})\,\mathbf{e}_2 + v^3(\mathbf{x})\,\mathbf{e}_3.$$

The three component functions v^1, v^2, v^3 are scalar-valued functions, but together they represent one geometric vector. If the coordinate axes are rotated, the components change in a coordinated way so that the physical vector remains the same.

This is why a vector should not be identified with a column of numbers alone. The column

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

represents a vector only after a basis has been specified. Relative to a different basis, the same vector generally has different components.

Consider the two-dimensional velocity field

$$\mathbf{v}(x, y) = -\Omega y\,\mathbf{e}_x + \Omega x\,\mathbf{e}_y,$$

where Ω is a constant with units of inverse time. At the point $(x, y) = (2\,\mathrm{m}, 1\,\mathrm{m})$,

$$\mathbf{v}(2\,\mathrm{m}, 1\,\mathrm{m}) = -\Omega(1\,\mathrm{m})\,\mathbf{e}_x + \Omega(2\,\mathrm{m})\,\mathbf{e}_y.$$

If $\Omega = 0.5\,\mathrm{s}^{-1}$, then

$$\mathbf{v} = -0.5\,\mathrm{m\,s^{-1}}\,\mathbf{e}_x + 1.0\,\mathrm{m\,s^{-1}}\,\mathbf{e}_y.$$

The local speed is the magnitude

$$\|\mathbf{v}\| = \sqrt{(-0.5)^2 + (1.0)^2}\,\mathrm{m\,s^{-1}} = \sqrt{1.25}\,\mathrm{m\,s^{-1}}.$$

This example shows how a vector field is used operationally: choose a point, evaluate each component there, and combine the components into a vector with magnitude and direction. The complete field may be visualized as an arrow attached to every point of the domain.

Tensor fields

A tensor field assigns a tensor to every point. Tensors generalize scalars and vectors. A scalar may be regarded as a tensor of rank zero, and a vector as a tensor of rank one. A rank-two tensor contains information about how one direction is related to another.

In coordinates, a rank-two tensor field may be written as an array of components

$$T^{ij}(\mathbf{x}) \quad \text{or} \quad T_{ij}(\mathbf{x}).$$

However, not every array of numbers is automatically a tensor. The defining property is the transformation law: when coordinates are changed, the components must transform so that the underlying geometric or physical object remains well defined.

A concrete example is the stress tensor $\boldsymbol{\sigma}$ in a continuous material. At each point, it determines the force per unit area acting on a small surface through that point. The force depends not only on position but also on the orientation of the surface. If $\mathbf{n}$ is a unit normal to the surface, the traction vector is

$$\mathbf{t} = \boldsymbol{\sigma}\,\mathbf{n}.$$

In two dimensions, suppose that at one point

$$\boldsymbol{\sigma} = \begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix} \text{MPa}.$$

For a surface whose unit normal is $\mathbf{n} = \mathbf{e}_x$, represented by

$$\mathbf{n} = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

we obtain

$$\mathbf{t} = \begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{MPa} = \begin{pmatrix} 4 \\ 1 \end{pmatrix} \text{MPa}.$$

For a surface with normal $\mathbf{e}_y$,

$$\mathbf{t} = \begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{MPa} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \text{MPa}.$$

The tensor therefore contains more information than a single vector: it tells us which traction vector is associated with each possible surface direction. This is a useful model for understanding why tensor fields naturally appear when physical effects depend on orientation.

Other important tensor fields include the electromagnetic field tensor, the energy-momentum tensor, and the spacetime metric. Each will be introduced only when there is a physical problem that requires it.

Transformation behaviour is the essential distinction

It is tempting to distinguish the three cases simply by counting components:

- one component for a scalar;
- several components for a vector;
- a matrix of components for a rank-two tensor.

This is useful as a first picture, but it is not the complete definition. The essential question is how the components change when the basis or coordinates change.

A scalar field obeys

$$\phi'(x') = \phi(x),$$

meaning that both expressions assign the same scalar to the same physical point. A vector transforms by one copy of the coordinate transformation matrix, while a rank-two tensor transforms by two copies. The precise formulas will be developed when tensor notation is introduced in Chapter 4.

For now, the practical lesson is:

> The components depend on the description; the scalar, vector, or tensor represented by those components is the physical or geometric object.

Several fields may describe one system

A physical system is rarely described by only one field. A fluid may require a mass-density field $\rho(\mathbf{x}, t)$, a pressure field $p(\mathbf{x}, t)$, a temperature field $T(\mathbf{x}, t)$, and a velocity field $\mathbf{v}(\mathbf{x}, t)$. These fields have different value types but share the same spatial and temporal domain.

Similarly, electrodynamics uses a scalar charge-density field, a vector current-density field, electric and magnetic vector fields, and later a single antisymmetric tensor field that combines the electric and magnetic information in a relativistic form.

The choice of field variables is part of constructing a theory. We must choose enough fields to describe the relevant physical information, but we should avoid introducing quantities that are redundant unless that redundancy serves a useful structural purpose.

What has been established

A scalar field assigns one coordinate-independent number to each point. A vector field assigns a geometric vector whose components depend on the chosen basis. A tensor field assigns a multilinear object that can relate several directions and whose components obey a definite transformation law.

The number of written components is not the deepest distinction. What matters is how the assigned object behaves under a change of coordinates. In the next section, we turn from the type of field value to the domain on which the field is defined and to the roles played by space and time.

1.3 Space-Time Dependence and Field Domains

A field is not specified completely until its domain is known. The formula

$$\phi(x, t) = A\cos(kx - \omega t)$$

may describe a disturbance on an infinite line, a wave on a finite string, or a local approximation inside a larger system. The same expression can represent different physical problems depending on the allowed values of x, the time interval under consideration, and the conditions imposed at the boundary.

The domain is therefore not a technical detail added after the field has been defined. It is part of the model.

Spatial domains

A field may be defined on spaces of different dimensions.

For a thin rod or string, one spatial coordinate may be sufficient:

$$D = [0, L] \subset \mathbb{R}.$$

A field on the rod is then written as $u(x, t)$, with $0 \leq x \leq L$. The reduction to one dimension is a modelling assumption. It means that variations across the thickness of the rod are neglected or averaged.

For a thin plate or membrane, the domain may be a region $D \subset \mathbb{R}^2$, and a field is written as

$$\phi = \phi(x, y, t).$$

For a field in ordinary three-dimensional space, the domain is a region $D \subset \mathbb{R}^3$, and we write

$$\phi = \phi(x, y, z, t) \quad \text{or} \quad \phi = \phi(\mathbf{x}, t).$$

The notation $\mathbf{x}$ is compact, but it should not hide the fact that three independent spatial coordinates are present.

Some fields live on curved surfaces or curved spaces. For example, the temperature on the surface of a sphere is naturally a field on that sphere, not on the entire surrounding three-dimensional space. Coordinates such as latitude and longitude may be used to label points, but the sphere itself is the geometric domain. Later, when gravity is discussed, spacetime will play this role as a curved domain.

Static and time-dependent fields

A *static field* does not change with time. It is written as

$$\phi = \phi(\mathbf{x}).$$

This does not mean that the field is spatially uniform. A static electric potential may vary strongly from point to point while remaining unchanged in time.

A *time-dependent field* is written as

$$\phi = \phi(\mathbf{x}, t).$$

At each fixed time $t = t_0$, the function $\phi(\mathbf{x}, t_0)$ is one field configuration. As time changes, the system passes through a family of configurations.

It is helpful to separate two related objects:

- the *configuration at time* t_0, which is a function of space only;
- the *field history*, which is a function of both space and time.

For a vibrating string, a photograph at one instant shows the configuration $u(x, t_0)$. A video of the motion represents the history $u(x, t)$.

Spacetime as a single domain

In nonrelativistic notation, space and time are usually written separately. In relativistic field theory, they are combined into a spacetime point. We write

$$x^\mu = (x^0, x^1, x^2, x^3),$$

where the time coordinate is commonly chosen as $x^0 = ct$ or, in units with $c = 1$, simply $x^0 = t$.

A relativistic field may then be written as

$$\phi = \phi(x^\mu).$$

This notation does not erase the physical distinction between time and space. It organizes them into one structure so that Lorentz transformations can be handled consistently. The geometric details will be developed in Chapter 4. At this stage, the important idea is that a field assigns data to events in spacetime, not merely to points in space.

Partial derivatives and what they measure

When a field depends on several independent variables, its rate of change is described by partial derivatives. Consider the field

$$u(x, t) = A\cos(kx)\cos(\omega t).$$

The spatial derivative is

$$\frac{\partial u}{\partial x} = -Ak \sin(kx) \cos(\omega t).$$

This derivative measures how the displacement changes when the position is varied while the time is held fixed. For a string, it is related to the local slope of the string.

The time derivative is

$$\frac{\partial u}{\partial t} = -A\omega \cos(kx) \sin(\omega t).$$

This derivative measures how the displacement at one fixed position changes with time. For a string, it is the transverse velocity of the material point labelled by x.

The two derivatives answer different physical questions. Confusing them would mean confusing a comparison between neighbouring points at one instant with a comparison between different instants at one point.

The dimensions provide a useful check. If u has dimensions of length, then

$$\left[\frac{\partial u}{\partial x}\right] = 1, \qquad \left[\frac{\partial u}{\partial t}\right] = \text{length/time}.$$

The spatial derivative is dimensionless in this example because it represents a slope, while the time derivative has the dimensions of velocity.

The field equation is not the whole problem

A differential equation usually admits many solutions. To determine the physical field, one must also specify the domain and appropriate data.

Consider a string occupying

$$0 \le x \le L$$

and satisfying the wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}.$$

This equation alone does not tell us which motion occurs. We also need initial data, such as

$$u(x,0) = f(x), \qquad \frac{\partial u}{\partial t}(x,0) = g(x).$$

The function $f(x)$ gives the initial shape of the string, and $g(x)$ gives the initial velocity of each point.

If the endpoints are fixed, we also impose boundary conditions:

$$u(0,t) = 0, \qquad u(L,t) = 0.$$

The complete mathematical problem therefore consists of four ingredients:

1. the field variable $u(x,t)$;
2. the domain $0 \le x \le L$;
3. the field equation;
4. the initial and boundary data.

Changing the boundary conditions changes the physical problem. A string with fixed endpoints, a string with one free endpoint, and a periodic string may obey the same local differential equation but have different allowed motions.

Boundary conditions encode physical information

A boundary condition is not merely a device for selecting one mathematical solution. It represents how the system interacts with its surroundings.

For a temperature field, fixing T at the boundary may represent contact with a heat reservoir. Fixing the normal heat flux may represent insulation or a prescribed flow of heat. For an electromagnetic field, boundary conditions describe conductors, interfaces between materials, incoming radiation, or behaviour at large distances.

It is therefore useful to ask what physical arrangement produces the chosen boundary condition. A mathematically convenient condition may not describe the experiment under consideration.

Local coordinates and physical points

Coordinates are labels assigned to points of the domain. The physical point is not identical with its coordinate tuple. On a flat Cartesian domain, the distinction may seem unnecessary because the coordinates are especially simple. On curved surfaces or in curvilinear coordinates, it becomes essential.

For example, the north pole of a sphere is one physical point even though longitude is not uniquely defined there. A field value at the north pole must be well defined independently of this coordinate ambiguity.

This observation prepares an important theme of field theory: equations should describe geometric or physical relations, not depend on an arbitrary choice of coordinates. Component formulas may change when coordinates are changed, while the underlying statement remains the same.

A checklist for specifying a field problem

Before manipulating a field formula, identify the following information:

1. What is the spatial domain?
2. Is the field static or time dependent?
3. What type of value is assigned to each point?
4. Which coordinates are being used?
5. What differential equation does the field satisfy?
6. What initial data are specified?
7. What boundary conditions are imposed?

If any of these items is missing, the problem may be incomplete or ambiguous.

What has been established

A field is defined on a specific spatial or spacetime domain. Its dependence on space describes variation from point to point, while its dependence on time describes the evolution of configurations. Partial derivatives express these distinct kinds of change.

A field equation must be interpreted together with its domain, initial data, and boundary conditions. The next section examines another structural property of field theories: locality, the idea that the behaviour at a point is governed by field values and derivatives in its immediate neighbourhood.

1.4 Locality, Sources, and Field Response

A field theory becomes especially useful when it replaces a direct interaction between distant objects by a local description. Instead of saying that one object acts immediately on another object far away, we introduce a field that exists throughout the intervening region. The source changes the field near itself, the field propagates or adjusts according to a differential equation, and another object responds to the field at its own position.

This viewpoint separates three questions:

1. What produces the field?
2. How does the field behave between sources?
3. How does matter respond to the field locally?

The answers are encoded in sources, field equations, and coupling laws.

What locality means

A field equation is called local when the behaviour at a point is determined by the field and a finite number of its derivatives evaluated at that point. A schematic local equation has the form

$$\mathcal{F}(\phi(x), \partial_\mu \phi(x), \partial_\mu \partial_\nu \phi(x), \ldots) = J(x),$$

where $J(x)$ represents a source.

The equation may involve neighbouring values indirectly through derivatives, but it does not require an independent instruction involving every distant point. A derivative is local because it is determined by how the field changes in an arbitrarily small neighbourhood.

For example, the one-dimensional wave equation

$$\frac{\partial^2 u}{\partial t^2}(x,t) = c^2 \frac{\partial^2 u}{\partial x^2}(x,t)$$

relates the acceleration of the string at (x,t) to the curvature of the string at the same point and time. The curvature is calculated from nearby points, not from the entire string at once.

Locality does not mean that distant regions never influence one another. It means that the influence is transmitted through successive local interactions. A disturbance created near one end of a string changes neighbouring parts of the string, and the disturbance then travels along it.

Sources

A source is a quantity that drives or generates a field. The source may be concentrated at isolated points, distributed throughout a region, or imposed at a boundary.

Examples include:

- electric charge density as a source of the electromagnetic field;
- mass-energy as a source of the gravitational field;
- an external force density as a source of displacement in an elastic medium;
- heat production as a source in a temperature equation.

A simple linear field equation may be written as

$$\mathcal{L}\phi = J,$$

where $\mathcal{L}$ is a differential operator, ϕ is the field, and J is the source.

This notation is compact, but each symbol has a different role. The operator tells us how the field responds, the source tells us what drives it, and the boundary or initial data select the physical solution.

A worked static example

Consider a scalar field $\phi(x)$ on the interval

$$0 \le x \le L.$$

Suppose that it satisfies

$$\frac{\mathrm{d}^2\phi}{\mathrm{d}x^2} = -\rho_0,$$

where ρ_0 is a constant source density. We impose the boundary conditions

$$\phi(0) = 0, \qquad \phi(L) = 0.$$

The equation can be solved by integrating twice. First,

$$\frac{\mathrm{d}\phi}{\mathrm{d}x} = -\rho_0 x + C_1.$$

Integrating again gives

$$\phi(x) = -\frac{\rho_0}{2}x^2 + C_1 x + C_2.$$

The first boundary condition gives

$$\phi(0) = C_2 = 0.$$

The second boundary condition gives

$$0 = -\frac{\rho_0}{2}L^2 + C_1 L,$$

so

$$C_1 = \frac{\rho_0 L}{2}.$$

The field is therefore

$$\boxed{\phi(x) = \frac{\rho_0}{2}x(L-x)}.$$

We can verify the result directly:

$$\frac{\mathrm{d}\phi}{\mathrm{d}x} = \frac{\rho_0}{2}(L-2x),$$

and

$$\frac{\mathrm{d}^2\phi}{\mathrm{d}x^2} = -\rho_0.$$

The solution also satisfies $\phi(0) = \phi(L) = 0$.

This example shows several important features of a field problem. The source fixes the local curvature of the field, while the boundary conditions determine the integration constants and therefore the global shape. Neither the source nor the boundary conditions alone are sufficient.

If $\rho_0 > 0$, the field is largest at the centre. Setting the derivative equal to zero,

$$\frac{\mathrm{d}\phi}{\mathrm{d}x} = 0,$$

gives

$$x = \frac{L}{2}.$$

At that point,

$$\phi\left(\frac{L}{2}\right) = \frac{\rho_0 L^2}{8}.$$

Thus the differential equation, the source, and the boundary conditions together determine not only the formula but also qualitative features such as where the maximum occurs.

Source-free does not mean field-free

In a region where the source vanishes, the field may satisfy a homogeneous equation,

$$\mathcal{L}\phi = 0.$$

This does not imply $\phi = 0$. A source located outside the region may produce a nonzero field inside it. Boundary data may also sustain a nonzero solution. Electromagnetic waves can travel through a region containing no charge or current.

The distinction is important:

- *source-free* means that the local source term vanishes;
- *field-free* means that the field itself vanishes.

These are different statements.

Static response and dynamical propagation

A static equation describes a field configuration after time dependence has been neglected. A dynamical equation describes how disturbances evolve.

A static source-field relation may have the form

$$\nabla^2\phi = -\rho.$$

A corresponding dynamical equation may have the form

$$\frac{\partial^2\phi}{\partial t^2} - c^2\nabla^2\phi = J.$$

The dynamical equation contains a propagation speed c. If the source changes, the response at a distant point does not appear immediately; it arrives after the disturbance has propagated.

Static solutions are still useful, but they must be interpreted correctly. They describe time-independent configurations or approximations in which propagation effects are negligible. A static formula should not automatically be read as evidence of instantaneous action at a distance.

Fields replace direct distant action

Newton's law of gravitation and Coulomb's law can be written as forces depending directly on the separation between two objects. Such formulas are effective and often sufficient for static problems. Field theory reorganizes the description.

For electromagnetism, the source produces an electromagnetic field. A test charge at position $\mathbf{x}$ responds to the field evaluated at that same position. The force law is local in this sense:

$$\mathbf{F}(t) = q\big[\mathbf{E}(\mathbf{x}(t), t) + \mathbf{v}(t) \times \mathbf{B}(\mathbf{x}(t), t)\big].$$

The charge does not need direct access to the current state of a distant source. It responds to the field where it is located. Maxwell's equations determine how changes in the sources affect the field and how those changes propagate.

This separation becomes essential when sources move rapidly or radiation is emitted. A direct static force law is then insufficient.

Linear response and superposition

If the field equation is linear,

$$\mathcal{L}\phi = J,$$

and two sources J_1 and J_2 produce fields ϕ_1 and ϕ_2, then

$$\mathcal{L}\phi_1 = J_1, \qquad \mathcal{L}\phi_2 = J_2.$$

By linearity,

$$\mathcal{L}(\phi_1 + \phi_2) = J_1 + J_2.$$

Thus the combined source produces the combined field

$$\phi = \phi_1 + \phi_2.$$

This is the principle of superposition. It allows complicated sources to be decomposed into simpler pieces. Later, Green functions will turn this principle into a systematic method of solution.

Nonlinear field equations do not generally satisfy superposition. In that case, two individual solutions cannot simply be added. Recognizing whether an equation is linear is therefore one of the first practical checks in a field problem.

Local laws and global solutions

A local differential equation constrains the field point by point, but its solution is a global object defined over the entire domain. The global solution depends on local laws together with initial and boundary data.

This distinction explains why local theories can produce rich large-scale behaviour. Waves, resonances, standing patterns, and long-range fields emerge from local differential equations applied across a domain.

The local equation tells us how neighbouring values are related. The domain and boundary conditions determine how those local relations fit together into one complete configuration.

What has been established

A local field theory describes the behaviour at a point using the field, its derivatives, and sources at that point. Distant influence is transmitted through the field rather than imposed as an independent instantaneous rule.

Sources drive fields, but source-free regions may still contain nonzero fields. Static equations describe configurations, while dynamical equations describe propagation. Linear equations admit superposition, which will later support powerful solution methods.

The next section examines why a field represents a system with continuously many degrees of freedom and how this statement should be understood without treating neighbouring field values as completely unrelated.

1.5 Degrees of Freedom in Continuous Systems

A degree of freedom is an independent quantity needed to specify the configuration of a system. For a particle moving on a line, one coordinate $q(t)$ is sufficient. For a particle moving in three-dimensional space, three coordinates are required. For N particles moving on a line, a configuration may be written as

$$\mathbf{q}(t) = \big(q_1(t), q_2(t), \ldots, q_N(t)\big).$$

The configuration is one point in an N-dimensional configuration space.

A continuous system is different. A string, fluid, elastic body, or field occupies a region containing continuously many points. To describe its configuration, we need a function rather than a finite list of numbers. The displacement of a string, for example, is written as

$$u = u(x,t), \qquad 0 \le x \le L.$$

At a fixed time, the complete configuration is the function $u(x)$. Informally, we say that there is one degree of freedom for each value of x. This statement is useful, but it must be interpreted carefully.

From finitely many coordinates to a field

Suppose that a string is approximated by N marked points. Let

$$x_n = n\,a, \qquad n = 0, 1, \ldots, N-1,$$

where

$$a = \frac{L}{N-1}$$

is the spacing between neighbouring points.

We record the displacement at each point:

$$u_n(t) = u(x_n, t).$$

The approximate configuration is then the finite vector

$$\mathbf{u}(t) = \big(u_0(t), u_1(t), \ldots, u_{N-1}(t)\big).$$

For $N = 5$, only five displacement values are stored. The shape between the marked points must be inferred by interpolation. If N is increased, more detail is retained. In the formal continuum limit,

$$N \longrightarrow \infty, \qquad a \longrightarrow 0, \qquad L \text{ fixed},$$

the finite list is replaced by the continuous function $u(x,t)$.

This is one practical meaning of a field: it is the limiting description obtained when a system is resolved at arbitrarily small spatial intervals.

A numerical example of increasing resolution

Take a string of length $L = 1\,\mathrm{m}$ with the configuration

$$u(x) = 0.02\,\mathrm{m}\,\sin\left(\frac{\pi x}{L}\right).$$

If we sample only the endpoints and the midpoint, then

$$x_0 = 0, \qquad x_1 = \frac{L}{2}, \qquad x_2 = L,$$

and

$$(u_0, u_1, u_2) = (0, 0.02\,\mathrm{m}, 0).$$

This three-number description captures the maximum displacement but not the detailed curvature.

With five equally spaced points,

$$x_n = \frac{nL}{4}, \qquad n = 0, 1, 2, 3, 4,$$

we obtain

$$\begin{aligned}
u_0 &= 0, \\
u_1 &= 0.02\sin\left(\frac{\pi}{4}\right)\mathrm{m}, \\
u_2 &= 0.02\,\mathrm{m}, \\
u_3 &= 0.02\sin\left(\frac{3\pi}{4}\right)\mathrm{m}, \\
u_4 &= 0.
\end{aligned}$$

Since

$$\sin\left(\frac{\pi}{4}\right) = \sin\left(\frac{3\pi}{4}\right) = \frac{1}{\sqrt{2}},$$

the sampled configuration is

$$\mathbf{u} = \left(0, \frac{0.02}{\sqrt{2}}, 0.02, \frac{0.02}{\sqrt{2}}, 0\right)\mathrm{m}.$$

The field formula contains all these finite approximations simultaneously. Any desired sampling resolution can be extracted from it.

The values are not arbitrary independent numbers

The phrase "one degree of freedom at each point" does not mean that neighbouring field values are always unrelated. Physical fields are usually required to be continuous or differentiable, and their equations relate values in nearby regions.

If the displacement of a string changed from 1 cm to 100 m over an arbitrarily small distance, the continuum model would usually cease to be physically meaningful. Smoothness assumptions restrict the allowed configurations.

A finite-difference approximation makes this relation visible. The spatial derivative at x_n may be approximated by

$$\frac{\partial u}{\partial x}(x_n, t) \approx \frac{u_{n+1}(t) - u_n(t)}{a}.$$

If the derivative must remain finite as $a \to 0$, then the difference $u_{n+1} - u_n$ must decrease with a. Neighbouring values cannot vary arbitrarily.

Boundary conditions impose further restrictions. For a string fixed at both ends,

$$u(0, t) = 0, \qquad u(L, t) = 0.$$

The endpoint values are not free degrees of freedom. Constraints and symmetries may similarly reduce the number of independent variables.

Configuration space becomes a function space

For a system with N coordinates, the configuration space may be $\mathbb{R}^N$. For a field, the configuration is a function, so the configuration space is a space of functions.

For a fixed string, one possible configuration space is schematically

$$\mathcal{C} = \{u : [0, L] \to \mathbb{R} \mid u(0) = u(L) = 0\},$$

with additional smoothness conditions when derivatives are needed.

The symbol $\mathcal{C}$ does not represent ordinary three-dimensional space. Each point of $\mathcal{C}$ represents one entire shape of the string. Moving through configuration space means changing from one complete field configuration to another.

This is why the action of a field theory is a functional. Its input is not a finite coordinate vector but a complete field history.

A field configuration versus a complete dynamical state

For many second-order dynamical equations, the field value at one instant is not enough to predict the future. We must also specify its time derivative.

For a string, the initial data are

$$u(x, 0) = f(x), \qquad \frac{\partial u}{\partial t}(x, 0) = g(x).$$

The function f gives the initial shape, while g gives the initial velocity distribution. Two strings may have the same shape at one instant but move differently because their velocity fields differ.

Thus we should distinguish:

- a *configuration*, specified by $u(x)$;
- a *dynamical state*, often specified by $u(x)$ together with $\partial u/\partial t$.

Later, in Hamiltonian field theory, the time derivative will be replaced by a conjugate momentum field.

Normal modes provide another set of coordinates

A field may also be described by expanding it in a set of spatial patterns. For a string fixed at $x = 0$ and $x = L$, a useful expansion is

$$u(x,t) = \sum_{n=1}^{\infty} q_n(t) \sin\left(\frac{n\pi x}{L}\right).$$

Each sine function already satisfies the endpoint conditions. The time-dependent coefficient $q_n(t)$ tells us how strongly the n-th normal mode is excited.

In this description, the field is represented not by one value at every point but by the infinite list

$$q_1(t), q_2(t), q_3(t), \ldots$$

The two descriptions contain the same information when the expansion is valid:

- position-space description: $u(x,t)$;
- mode description: the coefficients $q_n(t)$.

For example, the configuration

$$u(x,0) = A \sin\left(\frac{\pi x}{L}\right) + B \sin\left(\frac{2\pi x}{L}\right)$$

has only two nonzero mode amplitudes:

$$q_1(0) = A, \qquad q_2(0) = B, \qquad q_n(0) = 0 \quad \text{for } n \geq 3.$$

The full system permits infinitely many modes, but a particular configuration may involve only a few of them.

This mode viewpoint is important in wave physics and later becomes central in quantum field theory. It also clarifies that "infinitely many degrees of freedom" can be organized into a sequence of ordinary time-dependent coordinates.

Continuum models are approximations with a scale of validity

Many classical fields describe matter that is microscopically discrete. A string is made of atoms, a fluid is made of molecules, and an elastic solid has a crystal or molecular structure. The continuum description is effective when the length scales of interest are much larger than the microscopic spacing.

If a wave has wavelength λ and the microscopic spacing is a, the continuum approximation is usually most reliable when

$$\lambda \gg a.$$

At wavelengths comparable with the microscopic scale, the discrete structure may become observable, and the continuum field equation may need correction.

A field theory can therefore be exact within a mathematical model while still being an approximation to a more microscopic physical system. The domain of validity should always be kept in mind.

Discretization as a computational method

The same idea used to motivate the continuum limit is also used in the opposite direction for numerical calculations. A continuous field is replaced by values on a finite grid, derivatives are approximated by finite differences, and the field equation becomes a large system of algebraic or ordinary differential equations.

Increasing the number of grid points should improve the approximation when the numerical method is stable and convergent. Thus the relation

$$\text{finite system} \longleftrightarrow \text{continuous field}$$

works in both directions:

- a dense discrete system motivates a continuum field;
- a continuum field is discretized for numerical solution.

What has been established

A field represents a system whose configuration requires a function rather than a finite list of coordinates. This can be understood by sampling the field at N points and taking the limit of increasing resolution.

The field values are not generally arbitrary because smoothness, field equations, constraints, and boundary conditions relate them. The infinite-dimensional configuration space can be described either through values in position space or through infinitely many mode amplitudes.

The next section makes the discrete-to-continuum passage explicit. Starting from a chain of masses connected by springs, we will derive the wave equation and the corresponding continuum energy.

1.6 From Discrete Oscillators to Continuous Media

The continuum description of a string can be derived from a discrete mechanical model. This derivation is important because it shows that a field equation is not an arbitrary formula. It may emerge from ordinary Newtonian mechanics when a system contains many closely spaced degrees of freedom.

We consider a chain of identical masses connected by identical springs. The masses are separated by a fixed equilibrium distance a. The transverse displacement of the n-th mass is denoted by

$$u_n(t).$$

The equilibrium position of that mass is

$$x_n = na.$$

The goal is to replace the discrete variables

$$u_0(t), u_1(t), u_2(t), \dots$$

by a continuous displacement field

$$u = u(x,t).$$

Assumptions of the model

The derivation uses several assumptions that should be stated explicitly.

1. Every mass has the same mass m.
2. Neighbouring masses are connected by identical springs with spring constant κ.
3. Only nearest neighbours interact.
4. The transverse displacements are small enough that the restoring force is linear.
5. The displacement changes smoothly over distances comparable with the spacing a.

The final continuum equation is valid only within the regime where these assumptions are reasonable.

The discrete elastic energy

For small relative displacements, the spring between masses n and $n+1$ stores the potential energy

$$V_{n,n+1} = \frac{\kappa}{2}\big(u_{n+1} - u_n\big)^2.$$

The two springs connected to the n-th mass contribute

$$V = \frac{\kappa}{2}\big(u_n - u_{n-1}\big)^2 + \frac{\kappa}{2}\big(u_{n+1} - u_n\big)^2 + \text{terms independent of } u_n.$$

The force on the n-th mass is

$$F_n = -\frac{\partial V}{\partial u_n}.$$

Differentiating the first spring term gives

$$\frac{\partial}{\partial u_n}\left[\frac{\kappa}{2}(u_n - u_{n-1})^2\right] = \kappa(u_n - u_{n-1}),$$

while the second gives

$$\frac{\partial}{\partial u_n}\left[\frac{\kappa}{2}(u_{n+1} - u_n)^2\right] = -\kappa(u_{n+1} - u_n).$$

Therefore

$$\begin{aligned} F_n &= -\kappa(u_n - u_{n-1}) + \kappa(u_{n+1} - u_n) \\ &= \kappa\big(u_{n+1} - 2u_n + u_{n-1}\big). \end{aligned}$$

Newton's second law gives the discrete equation of motion

$$\boxed{m\frac{\mathrm{d}^2 u_n}{\mathrm{d}t^2} = \kappa\big(u_{n+1} - 2u_n + u_{n-1}\big)}.$$

The combination

$$u_{n+1} - 2u_n + u_{n-1}$$

is the discrete second difference. It measures whether the displacement of the n-th mass lies above or below the average of its neighbours.

If

$$u_n = \frac{u_{n-1} + u_{n+1}}{2},$$

then the second difference vanishes, and the two neighbouring springs exert no net transverse restoring force. If u_n lies above the average, the net force points downward. The force therefore tends to reduce local curvature.

Introducing a continuous field

We now assume that the discrete displacements are samples of a smooth function:

$$u_n(t) = u(x_n, t), \qquad x_n = na.$$

The neighbouring displacements are

$$u_{n+1}(t) = u(x + a, t), \qquad u_{n-1}(t) = u(x - a, t),$$

where $x = x_n$.

For a sufficiently smooth field, Taylor expansion about x gives

$$u(x + a, t) = u(x, t) + a\frac{\partial u}{\partial x} + \frac{a^2}{2}\frac{\partial^2 u}{\partial x^2} + \frac{a^3}{6}\frac{\partial^3 u}{\partial x^3} + O(a^4),$$
$$u(x - a, t) = u(x, t) - a\frac{\partial u}{\partial x} + \frac{a^2}{2}\frac{\partial^2 u}{\partial x^2} - \frac{a^3}{6}\frac{\partial^3 u}{\partial x^3} + O(a^4).$$

Adding these expressions and subtracting $2u(x, t)$, the terms linear and cubic in a cancel:

$$u(x + a, t) - 2u(x, t) + u(x - a, t) = a^2\frac{\partial^2 u}{\partial x^2} + O(a^4).$$

To leading order,

$$u_{n+1} - 2u_n + u_{n-1} \approx a^2\frac{\partial^2 u}{\partial x^2}.$$

The discrete second difference becomes the continuous second derivative.

Substituting this approximation into the discrete equation gives

$$m\frac{\partial^2 u}{\partial t^2} = \kappa a^2\frac{\partial^2 u}{\partial x^2}.$$

Identifying continuum parameters

The mass per unit length is

$$\mu = \frac{m}{a}.$$

We also define

$$T = \kappa a.$$

The quantity T has the dimensions of force:

$$[T] = [\kappa][a] = \frac{\mathrm{N}}{\mathrm{m}}\,\mathrm{m} = \mathrm{N}.$$

In the continuum string model, T plays the role of the tension.

Using

$$m = \mu a, \qquad \kappa = \frac{T}{a},$$

the discrete approximation becomes

$$\mu a \frac{\partial^2 u}{\partial t^2} = \frac{T}{a} a^2 \frac{\partial^2 u}{\partial x^2}.$$

Cancelling the common factor a, we obtain

$$\boxed{\mu \frac{\partial^2 u}{\partial t^2} = T \frac{\partial^2 u}{\partial x^2}}.$$

Dividing by μ,

$$\frac{\partial^2 u}{\partial t^2} = \frac{T}{\mu} \frac{\partial^2 u}{\partial x^2}.$$

Defining

$$c^2 = \frac{T}{\mu},$$

we arrive at the one-dimensional wave equation

$$\boxed{\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}}.$$

Dimensional check

The ratio T/μ has dimensions

$$\left[\frac{T}{\mu}\right] = \frac{\mathrm{kg\,m\,s^{-2}}}{\mathrm{kg\,m^{-1}}} = \mathrm{m^2\,s^{-2}}.$$

Therefore

$$[c] = \mathrm{m\,s^{-1}},$$

so c is a speed. This is consistent with its interpretation as the propagation speed of small disturbances along the string.

The dimensions of both sides of the wave equation also agree:

$$\left[\frac{\partial^2 u}{\partial t^2}\right] = \mathrm{m\,s^{-2}},$$

and

$$\left[c^2 \frac{\partial^2 u}{\partial x^2}\right] = \mathrm{m^2\,s^{-2}} \cdot \mathrm{m^{-1}} = \mathrm{m\,s^{-2}}.$$

Physical interpretation of the wave equation

The left-hand side,

$$\frac{\partial^2 u}{\partial t^2},$$

is the local transverse acceleration of the string.

The right-hand side contains

$$\frac{\partial^2 u}{\partial x^2},$$

which measures local curvature. A straight segment has zero second derivative and therefore no net transverse restoring force. A curved segment is accelerated toward a straighter configuration.

The coefficient T/μ compares the restoring effect of tension with the inertia of the string. Increasing the tension increases the propagation speed, while increasing the mass density decreases it:

$$c = \sqrt{\frac{T}{\mu}}.$$

The equation therefore has a direct mechanical interpretation. It is not merely a formal relation between derivatives.

Discrete waves and the long-wavelength limit

The continuum approximation can be checked by solving the discrete model with a wave-like ansatz. Let

$$u_n(t) = Ae^{i(kna-\omega t)}.$$

Then

$$\frac{\mathrm{d}^2 u_n}{\mathrm{d}t^2} = -\omega^2 u_n,$$

and

$$u_{n+1} = e^{ika}u_n, \qquad u_{n-1} = e^{-ika}u_n.$$

Substitution into the discrete equation gives

$$-m\omega^2 u_n = \kappa(e^{ika} - 2 + e^{-ika})u_n.$$

Using

$$e^{ika} + e^{-ika} = 2\cos(ka),$$

we find

$$-m\omega^2 = 2\kappa(\cos(ka) - 1).$$

Since

$$1 - \cos(ka) = 2\sin^2\left(\frac{ka}{2}\right),$$

the exact discrete dispersion relation is

$$\boxed{\omega^2 = \frac{4\kappa}{m}\sin^2\left(\frac{ka}{2}\right)}.$$

For long wavelengths,

$$ka \ll 1,$$

and

$$\sin\left(\frac{ka}{2}\right) \approx \frac{ka}{2}.$$

Therefore

$$\omega^2 \approx \frac{4\kappa}{m}\frac{k^2a^2}{4} = \frac{\kappa a^2}{m}k^2.$$

Because

$$\frac{\kappa a^2}{m} = \frac{T}{\mu} = c^2,$$

we obtain

$$\boxed{\omega^2 \approx c^2k^2}.$$

This is the dispersion relation of the continuum wave equation. The derivation shows exactly when the continuum model is valid: the wavelength must be much larger than the lattice spacing,

$$\lambda \gg a.$$

At shorter wavelengths, the sine function in the discrete dispersion relation cannot be replaced by its linear approximation, and the discrete structure becomes physically relevant.

From discrete energy to field energy

The total energy of the discrete chain is

$$E = \sum_n \left[\frac{m}{2}\dot{u}_n^2 + \frac{\kappa}{2}(u_{n+1} - u_n)^2\right].$$

In the continuum approximation,

$$\dot{u}_n \approx \frac{\partial u}{\partial t},$$

and

$$u_{n+1} - u_n \approx a\frac{\partial u}{\partial x}.$$

Using $m = \mu a$ and $\kappa = T/a$, the energy associated with one lattice interval becomes

$$\frac{m}{2}\dot{u}_n^2 + \frac{\kappa}{2}(u_{n+1} - u_n)^2 \approx a\left[\frac{\mu}{2}\left(\frac{\partial u}{\partial t}\right)^2 + \frac{T}{2}\left(\frac{\partial u}{\partial x}\right)^2\right].$$

The sum therefore becomes a Riemann sum:

$$E \approx \sum_n a\left[\frac{\mu}{2}\left(\frac{\partial u}{\partial t}\right)^2 + \frac{T}{2}\left(\frac{\partial u}{\partial x}\right)^2\right].$$

In the continuum limit,

$$\boxed{E = \int_0^L \left[\frac{\mu}{2}\left(\frac{\partial u}{\partial t}\right)^2 + \frac{T}{2}\left(\frac{\partial u}{\partial x}\right)^2\right] \mathrm{d}x}.$$

The first term is kinetic energy density, and the second is elastic energy density. The energy is no longer assigned to individual masses and springs; it is distributed continuously along the string.

This expression will later help motivate a Lagrangian density. The transition from a mechanical sum to a spatial integral is one of the basic structural steps from particle mechanics to field theory.

What the continuum limit keeps and what it forgets

The continuum field retains the long-wavelength collective motion of the discrete chain. It forgets details that depend on the individual masses and the precise lattice spacing.

The continuum approximation is therefore not obtained by simply replacing an index by a coordinate. Three coordinated replacements occur:

1. the discrete displacement $u_n(t)$ becomes the field $u(x,t)$;
2. finite differences become spatial derivatives;
3. sums of local contributions become spatial integrals.

The parameters must also be rescaled so that finite continuum quantities such as $\mu = m/a$ and $T = \kappa a$ remain meaningful.

What has been established

Starting from Newton's law for a chain of coupled oscillators, we derived the wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \qquad c = \sqrt{\frac{T}{\mu}}.$$

The derivation displayed the physical meaning of each term, the dimensional consistency of the result, and the assumptions behind the continuum approximation. The discrete dispersion relation showed that the field description is the long-wavelength limit of the microscopic model.

We also derived the continuum energy as an integral of an energy density. This prepares the later transition from mechanical Lagrangians to Lagrangian densities. The final section of the chapter now compares several physical fields and applies a common set of questions to each of them.

1.7 Basic Physical Examples of Fields

The concept of a field is useful because the same mathematical structure appears in many different physical systems. The symbols and units change, but the underlying questions remain similar:

1. What is the domain of the field?
2. What type of value is attached to each point?
3. Is the field static or time dependent?
4. Is it prescribed, or must it be solved for?
5. What produces it, and how is it measured?

This section applies these questions to several important examples. The purpose is not to develop the complete theory of each field. It is to learn how to recognize the mathematical and physical role played by a field.

Temperature as a scalar field

A temperature field assigns one real number to every point of a material body:

$$T = T(\mathbf{x}, t).$$

Its domain is the body together with the time interval under consideration. Its value is a scalar measured in kelvin.

Consider a thin rod of length L with the model temperature

$$T(x,t) = T_{\text{eq}} + \Delta T\, e^{-t/\tau} \cos\left(\frac{\pi x}{L}\right).$$

Here T_{eq} is an equilibrium temperature, ΔT is the initial temperature variation, and τ is a characteristic decay time.

At the left end,

$$T(0,t) = T_{\text{eq}} + \Delta T e^{-t/\tau}.$$

At the midpoint,

$$T\left(\frac{L}{2}, t\right) = T_{\text{eq}} + \Delta T e^{-t/\tau} \cos\left(\frac{\pi}{2}\right) = T_{\text{eq}}.$$

At the right end,

$$T(L,t) = T_{\text{eq}} - \Delta T e^{-t/\tau}.$$

The formula therefore represents a rod whose left and right ends initially lie above and below equilibrium by equal amounts. The exponential factor causes the variation to decrease with time.

The field can be sampled with thermometers, but a finite set of measurements gives only an approximation to the complete field. A theoretical temperature field interpolates between measurements and predicts values at unmeasured points.

The spatial gradient

$$\nabla T$$

points in the direction of the most rapid temperature increase. In heat conduction, this gradient helps determine the local heat flux. Thus derivatives of the field have direct physical meaning.

Density and velocity fields in a fluid

A fluid requires more than one field. Its mass density is a scalar field,

$$\rho = \rho(\mathbf{x}, t),$$

while its velocity is a vector field,

$$\mathbf{v} = \mathbf{v}(\mathbf{x}, t).$$

The density tells us how much mass is stored locally. If V is a region of space, the total mass inside it is

$$M_V(t) = \int_V ho(\mathbf{x}, t)\, \mathrm{d}^3 x.$$

For a uniform density ρ_0 inside a rectangular box of side lengths a, b, and c,

$$M = \int_0^a \int_0^b \int_0^c \rho_0 \, \mathrm{d}z \, \mathrm{d}y \, \mathrm{d}x = \rho_0 abc.$$

This calculation shows how a local field quantity produces a global observable by integration.

The velocity field gives the velocity of the fluid element passing through each point. As an example, consider

$$\mathbf{v}(x, y) = \alpha x \, \mathbf{e}_x - \alpha y \, \mathbf{e}_y,$$

where α has units of inverse time.

At $(x, y) = (2\,\mathrm{m}, 1\,\mathrm{m})$,

$$\mathbf{v} = 2\alpha \, \mathrm{m} \, \mathbf{e}_x - \alpha \, \mathrm{m} \, \mathbf{e}_y.$$

The flow stretches material in the x-direction and compresses it in the y-direction. The divergence is

$$\nabla \cdot \mathbf{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = \alpha - \alpha = 0.$$

The vanishing divergence indicates that local expansion in one direction is balanced by compression in the other. This is an example of how a derivative of a vector field describes a local property of a flow.

A complete fluid model may also contain pressure, temperature, and concentration fields. These fields interact through local differential equations.

Displacement of a string

The transverse displacement of a string is a scalar field on a one-dimensional spatial domain:

$$u = u(x, t), \qquad 0 \leq x \leq L.$$

Although the physical string lies in space, the coordinate x labels position along its equilibrium length, and u records displacement in a chosen transverse direction.

A travelling wave may be written as

$$u(x, t) = A \cos(kx - \omega t).$$

Its local transverse velocity is

$$\frac{\partial u}{\partial t} = A\omega \sin(kx - \omega t),$$

and its local slope is

$$\frac{\partial u}{\partial x} = -Ak \sin(kx - \omega t).$$

The two quantities are proportional for this right-moving wave:

$$\frac{\partial u}{\partial t} = -\frac{\omega}{k} \frac{\partial u}{\partial x}.$$

If the dispersion relation is $\omega = ck$, then

$$\frac{\partial u}{\partial t} = -c \frac{\partial u}{\partial x}.$$

This relation expresses the rigid translation of the wave profile to the right with speed c. A local time change is linked to the local spatial slope because the same shape is moving through the coordinate system.

The string example also shows that a scalar field need not represent a scalar quantity in all of three-dimensional space. The field value is scalar relative to the reduced one-dimensional model because only one transverse component has been retained.

Electric potential and electric field

Electrostatics uses both a scalar field and a vector field. The electric potential is a scalar field

$$\Phi = \Phi(\mathbf{x}),$$

while the electric field is a vector field

$$\mathbf{E} = \mathbf{E}(\mathbf{x}).$$

They are related by

$$\mathbf{E} = -\nabla\Phi.$$

For a point charge Q at the origin, the electrostatic potential in vacuum is

$$\Phi(r) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r},$$

where $r = \|\mathbf{x}\|$.

Because the potential depends only on the radial coordinate, its gradient is

$$\nabla\Phi = \frac{\mathrm{d}\Phi}{\mathrm{d}r}\,\widehat{\mathbf{r}}.$$

Differentiating,

$$\frac{\mathrm{d}\Phi}{\mathrm{d}r} = -\frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}.$$

Therefore

$$\boxed{\mathbf{E}(\mathbf{x}) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\widehat{\mathbf{r}}}.$$

For $Q > 0$, the field points radially outward. For $Q < 0$, it points inward.

This calculation illustrates how a vector field can be generated from the spatial variation of a scalar field. The potential describes an energy-like quantity per unit charge, while its gradient determines the local force direction.

A test charge q placed at $\mathbf{x}$ experiences

$$\mathbf{F} = q\mathbf{E}(\mathbf{x}).$$

The response is local: the charge responds to the field evaluated at its own position.

The electromagnetic field

When electric and magnetic effects vary with time, the theory requires the electric vector field $\mathbf{E}(\mathbf{x}, t)$ and magnetic vector field $\mathbf{B}(\mathbf{x}, t)$. These fields are coupled: a changing magnetic field is related to the circulation of the electric field, and a changing electric field contributes to the magnetic field.

In relativistic notation, $\mathbf{E}$ and $\mathbf{B}$ are not independent geometric objects. They are components of one antisymmetric rank-two tensor field,

$$F_{\mu\nu}(x).$$

Different inertial observers may divide the same tensor into electric and magnetic components differently. This is one reason tensor notation is not merely a compressed way of writing equations; it reveals which quantities belong to one relativistic field.

The detailed construction of $F_{\mu\nu}$ from the electromagnetic potential will be carried out in Chapter 7.

Gravitational potential and gravitational field

In Newtonian gravity, a mass distribution produces a scalar potential Φ_g. For a point mass M,

$$\Phi_g(r) = -\frac{GM}{r}.$$

The gravitational acceleration field is

$$\mathbf{g} = -\nabla\Phi_g.$$

Since

$$\frac{\mathrm{d}\Phi_g}{\mathrm{d}r} = \frac{GM}{r^2},$$

we obtain

$$\boxed{\mathbf{g} = -\frac{GM}{r^2}\hat{\mathbf{r}}}.$$

The minus sign indicates that the field points toward the mass.

The mathematical resemblance to electrostatics is clear. Both theories use a scalar potential whose gradient produces a force field. The physical interpretation differs: electric charges can have either sign, while ordinary mass is positive and Newtonian gravity is attractive.

General relativity replaces the Newtonian gravitational field by a dynamical spacetime metric tensor. The metric determines intervals, motion, and curvature. The final chapter will outline this geometric field description.

Stress as a tensor field

In a continuous material, the force acting across an internal surface depends on the orientation of that surface. A single force vector at each point is therefore insufficient. The stress tensor field

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}(\mathbf{x}, t)$$

assigns a linear map to each point. Given a unit normal $\mathbf{n}$, it produces the traction vector

$$\mathbf{t}(\mathbf{n}) = \boldsymbol{\sigma}\mathbf{n}.$$

The field value is a tensor because it relates one direction, the surface normal, to another vector, the force per unit area. The same tensor describes the traction on every possible surface orientation through the point.

This example demonstrates why tensors arise naturally in continuum physics. They encode directional relationships that cannot be represented by one scalar or one vector.

One system may require several coupled fields

Consider a heated fluid moving through a pipe. A useful model may require

$$\rho(\mathbf{x}, t), \qquad p(\mathbf{x}, t), \qquad T(\mathbf{x}, t), \qquad \mathbf{v}(\mathbf{x}, t).$$

The density ρ, pressure p, and temperature T are scalar fields. The velocity $\mathbf{v}$ is a vector field.

These fields are not separate descriptions of unrelated phenomena. The density and velocity are connected by mass conservation. Pressure gradients influence acceleration. Temperature may change density and pressure. Motion transports heat.

A field theory often consists of a coupled system of equations rather than one equation for one field. The first task is to identify the required fields and the physical principles that connect them.

A practical field-identification procedure

When a new field appears, analyze it in the following order.

1. **Name the field variable.** For example, T, ρ, $\mathbf{v}$, or $F_{\mu\nu}$.
2. **State the domain.** Is it a line, surface, spatial region, or spacetime?
3. **Identify the value type.** Is it scalar, vector, or tensor?
4. **Give the physical units.**
5. **Decide whether it is prescribed or dynamical.**
6. **Identify sources and boundary data.**
7. **State how the field is measured or how it affects matter.**
8. **Identify the relevant derivatives or integrals.**

This procedure prevents a field from becoming an unexplained symbol. It connects the mathematical object with the physical problem it is meant to describe.

Chapter summary

This chapter established the basic language of classical field theory.

A field assigns physical or mathematical information to every point of a domain. Depending on the value assigned, the field may be scalar, vector, or tensor. The classification is determined by transformation behaviour, not merely by the number of written components.

A field may depend on space and time. Its physical problem is not specified by a differential equation alone; the domain, initial data, and boundary conditions are also required. Local field equations relate values and derivatives at the same point, while sources drive the field and local coupling laws determine how matter responds.

Continuous systems contain infinitely many degrees of freedom in the sense that a configuration is a function rather than a finite coordinate vector. This structure can be understood by discretizing space or by expanding the field in infinitely many normal modes.

Most importantly, the wave equation was derived from a chain of coupled oscillators. Finite differences became derivatives, sums became integrals, and microscopic parameters combined into the continuum quantities μ and T. The derivation showed how a local field equation and a field energy emerge from ordinary mechanics.

The next chapter develops the action principle for particles. That may appear to be a temporary return from fields to mechanics, but it provides the variational language needed to derive field equations systematically. Once the mechanical argument is understood in detail, it can be extended from finitely many coordinates to an entire field.

Chapter 2

Action Principles: From Particles to Fields

2.1 Action as a Functional

The equations of mechanics are often introduced through forces. One specifies the forces acting on a particle and then uses Newton's second law to determine its acceleration. The action principle organizes the same dynamics in a different way. Instead of asking for the force at each instant, it assigns one number to an entire possible history of the system and then identifies the physical history through a stationarity condition.

This change of viewpoint is essential for field theory. A field configuration is already a function of space and time, so the natural dynamical object is not an ordinary function of a few variables but a *functional*: an object whose input is itself a function.

From instantaneous states to complete trajectories

Consider a particle moving on a line. Its position is described by a function

$$q = q(t), \qquad t_1 \le t \le t_2.$$

At one instant, the value $q(t)$ tells us where the particle is. The complete function q tells us the whole trajectory between the initial time t_1 and the final time t_2.

It is important to distinguish these two kinds of input:

- an ordinary function may take a number as input, for example $V(q)$;
- a functional takes an entire function as input, for example $S[q]$.

The square brackets in $S[q]$ emphasize that the argument is the complete trajectory, not merely the value of the coordinate at one time.

For example, the expression

$$F[q] = \int_{t_1}^{t_2} q(t)\,\mathrm{d}t$$

is a functional. Once a complete function $q(t)$ is supplied, the integral produces one number. If

$$q_1(t) = t, \qquad 0 \le t \le 1,$$

then

$$F[q_1] = \int_0^1 t\,\mathrm{d}t = \frac{1}{2}.$$

For a different trajectory,

$$q_2(t) = t^2, \qquad 0 \le t \le 1,$$

we obtain

$$F[q_2] = \int_0^1 t^2\,\mathrm{d}t = \frac{1}{3}.$$

The output is one number in each case, but that number depends on the entire shape of the input function over the interval.

The Lagrangian

In ordinary mechanics, the dynamical information is encoded in a function called the Lagrangian. For one generalized coordinate, it is usually written as

$$L = L(q, \dot{q}, t),$$

where

$$\dot{q} = \frac{\mathrm{d}q}{\mathrm{d}t}.$$

For a large class of mechanical systems,

$$L = T - V,$$

where T is the kinetic energy and V is the potential energy.

For a particle of mass m moving in a potential $V(q)$, the Lagrangian is

$$L(q, \dot{q}) = \frac{m}{2}\dot{q}^2 - V(q).$$

The Lagrangian is an ordinary function. At a specified time, it can be evaluated once the instantaneous position and velocity are known. Along a complete trajectory $q(t)$, however, it becomes a function of time:

$$t \longmapsto L(q(t), \dot{q}(t), t).$$

The action is obtained by integrating this quantity over the time interval.

Definition of the action

The action functional is defined by

$$\boxed{S[q] = \int_{t_1}^{t_2} L(q(t), \dot{q}(t), t)\, \mathrm{d}t}.$$

The input of S is a trajectory $q(t)$. The output is a single number. Different candidate trajectories generally produce different values of the action.

The action depends not only on the geometric curve traced by the particle but also on how the trajectory is parametrized in time, because the velocity $\dot{q}(t)$ enters the Lagrangian.

The endpoints and the time interval are part of the variational problem. When we later compare nearby trajectories, we must state whether the initial and final times are fixed and whether the endpoint positions are fixed. These choices determine which variations are allowed.

Dimensions of the action

Because the Lagrangian has the dimensions of energy,

$$[L] = \mathrm{J},$$

the action has dimensions

$$[S] = [L][t] = \mathrm{J\,s}.$$

In base SI units,

$$[S] = \mathrm{kg\,m^2\,s^{-1}}.$$

This dimensional check is useful. Every term in a Lagrangian must have the dimensions of energy, and every contribution to the action must have the dimensions of energy multiplied by time.

The same dimensions are carried by Planck's constant $\hbar$. Classical mechanics does not require $\hbar$, but the comparison becomes important in quantum mechanics, where the dimensionless ratio $S/\hbar$ appears naturally.

Example: a free particle along a prescribed path

For a free particle moving on a line,

$$V(q) = 0,$$

so the Lagrangian is

$$L = \frac{m}{2}\dot{q}^2.$$

Consider the prescribed linear trajectory

$$q(t) = q_1 + \frac{q_2 - q_1}{t_2 - t_1}(t - t_1).$$

Its velocity is constant:

$$\dot{q}(t) = \frac{q_2 - q_1}{t_2 - t_1}.$$

Substitution into the action gives

$$\begin{aligned} S[q] &= \int_{t_1}^{t_2} \frac{m}{2}\left(\frac{q_2 - q_1}{t_2 - t_1}\right)^2 \mathrm{d}t \\ &= \frac{m}{2}\left(\frac{q_2 - q_1}{t_2 - t_1}\right)^2 (t_2 - t_1) \\ &= \boxed{\frac{m(q_2 - q_1)^2}{2(t_2 - t_1)}}. \end{aligned}$$

This calculation does not yet prove that the linear trajectory is physical. It only evaluates the action for one candidate path. To identify the physical trajectory, we must compare it with nearby paths and determine whether the first-order change of the action vanishes.

Example: action of a trial trajectory

To see explicitly that the action depends on the whole path, consider a free particle on the interval

$$0 \le t \le T$$

with fixed endpoints

$$q(0) = 0, \qquad q(T) = a.$$

One admissible path is the linear trajectory

$$q_0(t) = \frac{a}{T}t.$$

A family of alternative paths with the same endpoints is

$$q_\varepsilon(t) = \frac{a}{T}t + \varepsilon \sin\left(\frac{\pi t}{T}\right),$$

because the sine term vanishes at both endpoints. The velocity is

$$\dot{q}_\varepsilon(t) = \frac{a}{T} + \varepsilon\frac{\pi}{T}\cos\left(\frac{\pi t}{T}\right).$$

For the free-particle Lagrangian,

$$S[q_\varepsilon] = \frac{m}{2}\int_0^T \dot{q}_\varepsilon^2(t)\,\mathrm{d}t.$$

Expanding the square,

$$\dot{q}_\varepsilon^2 = \frac{a^2}{T^2} + 2\varepsilon\frac{a\pi}{T^2}\cos\left(\frac{\pi t}{T}\right) + \varepsilon^2\frac{\pi^2}{T^2}\cos^2\left(\frac{\pi t}{T}\right).$$

The term linear in ε integrates to zero because

$$\int_0^T \cos\left(\frac{\pi t}{T}\right)\,\mathrm{d}t = 0.$$

Also,

$$\int_0^T \cos^2\left(\frac{\pi t}{T}\right)\,\mathrm{d}t = \frac{T}{2}.$$

Therefore

$$\boxed{S[q_\varepsilon] = \frac{ma^2}{2T} + \frac{m\pi^2}{4T}\varepsilon^2}.$$

The action changes only at second order in ε near $\varepsilon = 0$. This is the simplest preview of stationarity: the first derivative of the action with respect to the deformation parameter vanishes at the linear path.

Stationary does not always mean minimal

The physical trajectory is characterized by a stationary action, not universally by the smallest possible action. In symbols, the condition is written

$$\delta S = 0.$$

This means that the first-order change of the action vanishes for all allowed infinitesimal variations of the trajectory. Depending on the system and the time interval, the stationary trajectory may correspond to a local minimum, a local maximum, or a saddle point of the action.

The phrase "principle of least action" is historically common, but "principle of stationary action" is more precise. The essential statement is not

$$S[q_{\text{physical}}] < S[q]$$

for every other path. The essential statement is that sufficiently small admissible changes of the physical path produce no first-order change in S.

Local dynamics from a global quantity

The action is an integral over an entire time interval, so it may appear to define dynamics globally. Nevertheless, when the action is stationary under arbitrary local deformations of the trajectory, the result is a differential equation that holds at every time.

This is one of the central structural facts of variational mechanics:

> A condition imposed on the action of a complete trajectory produces a local equation of motion.

The mechanism behind this statement will be developed in the next sections. We will introduce a varied trajectory, calculate the first variation of the action, use integration by parts, control the boundary term, and derive the Euler–Lagrange equation.

Why this formulation is useful for field theory

For a particle, the action is a functional of a trajectory $q(t)$. For a field, the action will be a functional of a field configuration $\phi(x)$ over spacetime:

$$S[\phi] = \int \mathcal{L}(\phi, \partial_\mu \phi, x)\, \mathrm{d}^4 x.$$

The mechanical Lagrangian L is replaced by a Lagrangian density $\mathcal{L}$, and the time integral is replaced by a spacetime integral. The conceptual structure remains the same:

1. choose a candidate history;
2. evaluate its action;
3. compare it with nearby admissible histories;
4. require the first-order change of the action to vanish.

This is why the variational principle is first developed for particles. The particle calculation contains, in a simpler setting, the same ideas that will later be applied to fields.

What has been established

A functional maps an entire function to a number. In mechanics, the action functional assigns a number to every admissible trajectory:

$$S[q] = \int_{t_1}^{t_2} L(q, \dot{q}, t)\, \mathrm{d}t.$$

The Lagrangian is evaluated locally along the path, while the action summarizes the complete history over the chosen time interval. The physical path is not selected by evaluating one instantaneous state; it is selected through the stationarity of the action under allowed variations.

The next section constructs those variations explicitly and explains how a one-parameter family of nearby trajectories is used to calculate the first variation of the action.

2.2 Variations of Particle Trajectories

The action principle compares one candidate trajectory with nearby trajectories. To perform this comparison systematically, we introduce a parameter that labels a family of paths. The derivative with respect to that parameter defines the *variation* of the trajectory.

This construction is the central technical step of variational mechanics. Ordinary differentiation measures how a quantity changes when time changes. A variation measures how a complete trajectory changes when we move from one candidate path to another.

A one-parameter family of paths

Let

$$q = q(t)$$

be a reference trajectory on the interval

$$t_1 \leq t \leq t_2.$$

We construct nearby trajectories by writing

$$\boxed{q_\varepsilon(t) = q(t) + \varepsilon\eta(t)}.$$

Here:

- ε is a real parameter;
- $\eta(t)$ is a chosen deformation function;
- $q_0(t) = q(t)$ is the reference trajectory.

For small ε, the path q_ε lies close to q. Positive and negative values of ε deform the path in opposite directions.

At a fixed time t, the number $\eta(t)$ tells us how strongly the trajectory is displaced at that instant. The entire function η specifies the shape of the deformation over the complete interval.

Definition of the variation

The variation of the trajectory is defined by differentiating the family q_ε with respect to ε and then setting $\varepsilon = 0$:

$$\delta q(t) := \left.\frac{\partial q_\varepsilon(t)}{\partial \varepsilon}\right|_{\varepsilon=0}.$$

For the family

$$q_\varepsilon(t) = q(t) + \varepsilon\eta(t),$$

we obtain

$$\boxed{\delta q(t) = \eta(t)}.$$

The symbol δ does not represent a new time derivative. It denotes differentiation in the space of trajectories.

The distinction is therefore

$$\frac{\mathrm{d}q}{\mathrm{d}t} \quad \text{versus} \quad \delta q.$$

The first quantity compares the values of one trajectory at nearby times. The second compares nearby trajectories at the same time.

Variation of the velocity

The velocity along the varied trajectory is

$$\dot{q}_\varepsilon(t) = \frac{\mathrm{d}}{\mathrm{d}t} q_\varepsilon(t).$$

Using

$$q_\varepsilon(t) = q(t) + \varepsilon\eta(t),$$

we find

$$\dot{q}_\varepsilon(t) = \dot{q}(t) + \varepsilon\dot{\eta}(t).$$

The variation of the velocity is therefore

$$\begin{aligned}\delta\dot{q}(t) &:= \left.\frac{\partial \dot{q}_\varepsilon(t)}{\partial\varepsilon}\right|_{\varepsilon=0} \\ &= \dot{\eta}(t).\end{aligned}$$

Since $\delta q = \eta$, this may be written as

$$\boxed{\delta\dot{q} = \frac{\mathrm{d}}{\mathrm{d}t}(\delta q)}.$$

Thus the variation and the time derivative commute for the class of smooth variations considered here:

$$\delta\left(\frac{\mathrm{d}q}{\mathrm{d}t}\right) = \frac{\mathrm{d}}{\mathrm{d}t}(\delta q).$$

This identity will be used repeatedly. It is the reason that varying a Lagrangian containing $\dot{q}$ produces a term involving the time derivative of δq.

A geometric picture

A trajectory can be represented as a curve in the (t, q)-plane. The family

$$q_\varepsilon(t) = q(t) + \varepsilon\eta(t)$$

is then a family of nearby curves. For each fixed value of t, changing ε moves vertically from one curve to another.

The reference path is the curve corresponding to $\varepsilon = 0$. The function $\eta(t)$ gives the tangent direction of the family in the space of curves.

This geometric picture explains why a variation is not one additional trajectory by itself. It is an infinitesimal direction in which the reference trajectory may be deformed.

Variations with fixed endpoints

In the standard derivation of the Euler–Lagrange equation, the initial and final positions are held fixed:

$$q_\varepsilon(t_1) = q(t_1), \qquad q_\varepsilon(t_2) = q(t_2).$$

Substituting

$$q_\varepsilon(t) = q(t) + \varepsilon\eta(t)$$

gives

$$q(t_1) + \varepsilon\eta(t_1) = q(t_1),$$

and

$$q(t_2) + \varepsilon\eta(t_2) = q(t_2).$$

For these relations to hold for all sufficiently small ε, we require

$$\boxed{\eta(t_1) = 0, \qquad \eta(t_2) = 0}.$$

Equivalently,

$$\delta q(t_1) = 0, \qquad \delta q(t_2) = 0.$$

The deformation is otherwise arbitrary in the interior of the interval. It may be concentrated in a small region, oscillatory, or spread over the entire path, provided it is sufficiently smooth and satisfies the endpoint conditions.

The role of these conditions will be examined in detail in the next section. They are not decorative assumptions: they determine the boundary term produced by integration by parts.

Example of an admissible deformation

Suppose that

$$0 \leq t \leq T$$

and the endpoints are fixed. A simple admissible deformation is

$$\eta(t) = A \sin\left(\frac{\pi t}{T}\right),$$

because

$$\eta(0) = 0, \qquad \eta(T) = 0.$$

The corresponding family of trajectories is

$$q_\varepsilon(t) = q(t) + \varepsilon A \sin\left(\frac{\pi t}{T}\right).$$

The deformation is largest near the middle of the interval and vanishes at the endpoints. Another admissible choice is

$$\eta(t) = At(T - t).$$

Again,

$$\eta(0) = 0, \qquad \eta(T) = 0.$$

These examples show that the allowed deformation is not unique. The stationarity condition must hold for every admissible $\eta(t)$, not merely for one convenient choice.

A localized variation

To understand why a variational principle produces a local differential equation, it is useful to consider a deformation that is nonzero only in a small part of the interval.

Let t_a and t_b satisfy

$$t_1 < t_a < t_b < t_2.$$

We may choose a smooth function $\eta(t)$ such that

$$\eta(t) = 0$$

outside the interval (t_a, t_b). The varied and reference trajectories then agree everywhere except in that small time region.

If the action is stationary under all such localized deformations, the coefficient multiplying $\eta(t)$ in the first variation must vanish at every interior time. This idea will later be formalized by the fundamental lemma of the calculus of variations.

The varied action

The action of the varied path is

$$S[q_\varepsilon] = \int_{t_1}^{t_2} L\big(q_\varepsilon(t), \dot{q}_\varepsilon(t), t\big)\,\mathrm{d}t.$$

Once the deformation function $\eta(t)$ is chosen, this expression is an ordinary function of the parameter ε:

$$\varepsilon \longmapsto S[q_\varepsilon].$$

The first variation of the action is defined by

$$\boxed{\delta S[q;\eta] := \frac{\mathrm{d}}{\mathrm{d}\varepsilon} S[q_\varepsilon]\bigg|_{\varepsilon=0}}.$$

The notation $\delta S[q;\eta]$ makes two dependences visible:

- the variation is evaluated at the reference trajectory q;
- its value depends on the chosen deformation direction η.

When the context is clear, this is abbreviated to δS.

First-order expansion of the Lagrangian

For small ε, the Lagrangian along the varied path may be expanded to first order:

$$\begin{aligned} L(q_\varepsilon, \dot{q}_\varepsilon, t) &= L(q + \varepsilon\eta, \dot{q} + \varepsilon\dot{\eta}, t) \\ &= L(q, \dot{q}, t) + \varepsilon\left(\frac{\partial L}{\partial q}\eta + \frac{\partial L}{\partial \dot{q}}\dot{\eta}\right) + O(\varepsilon^2). \end{aligned}$$

Integrating over time gives

$$S[q_\varepsilon] = S[q] + \varepsilon \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q}\eta + \frac{\partial L}{\partial \dot{q}}\dot{\eta}\right)\mathrm{d}t + O(\varepsilon^2).$$

Therefore

$$\boxed{\delta S=\int_{t_1}^{t_2}\left(\frac{\partial L}{\partial q}\,\delta q+\frac{\partial L}{\partial \dot q}\,\delta \dot q\right)\,\mathrm{d}t}.$$

Using $\delta q=\eta$ and $\delta\dot q=\dot\eta$, we may also write

$$\delta S=\int_{t_1}^{t_2}\left(\frac{\partial L}{\partial q}\eta+\frac{\partial L}{\partial \dot q}\dot\eta\right)\,\mathrm{d}t.$$

At this stage, the derivative $\dot\eta$ still appears. To obtain a condition involving the arbitrary function η itself, we will transfer the time derivative from η to $\partial L/\partial\dot q$ by integration by parts.

Linearity of the first variation

The first variation is linear in the deformation direction. If

$$\eta(t)=a\eta_1(t)+b\eta_2(t),$$

then

$$\delta S[q;a\eta_1+b\eta_2]=a\,\delta S[q;\eta_1]+b\,\delta S[q;\eta_2].$$

This follows directly from the first-order expression for δS. Linearity is important because the first variation is the functional analogue of a directional derivative.

The second-order and higher-order changes of the action are generally not linear in η. They become relevant when one asks whether a stationary path is a minimum, maximum, or saddle point.

Several generalized coordinates

For a system described by coordinates

$$q^1(t),q^2(t),\dots,q^N(t),$$

we introduce one variation for each coordinate:

$$q^i_\varepsilon(t)=q^i(t)+\varepsilon\eta^i(t).$$

Then

$$\delta q^i=\eta^i,\qquad \delta\dot q^i=\dot\eta^i.$$

The first variation becomes

$$\delta S=\int_{t_1}^{t_2}\sum_{i=1}^{N}\left(\frac{\partial L}{\partial q^i}\eta^i+\frac{\partial L}{\partial \dot q^i}\dot\eta^i\right)\,\mathrm{d}t.$$

Each deformation function $\eta^i(t)$ may be chosen independently when the coordinates are independent. This independence will later produce one Euler–Lagrange equation for each generalized coordinate.

Common mistakes

Several distinctions must be kept clear.

1. The variation δq is not the same as a small time increment $\mathrm{d}q$. It compares different paths at the same time.

2. The parameter ε is not physical time. It labels trajectories in a family.

3. The function $\eta(t)$ is not chosen after the calculation to force a desired result. Stationarity must hold for every admissible η.
4. If the endpoints are fixed, the variation must vanish there. This does not imply that $\dot{\eta}$ also vanishes at the endpoints.
5. The equality $\delta\dot{q} = \mathrm{d}(\delta q)/\mathrm{d}t$ must be used explicitly when varying a velocity-dependent Lagrangian.

These points prevent the variational calculation from becoming a formal manipulation without a clear meaning.

Preparation for field variations

The same construction extends directly to a field. A reference field $\phi(x)$ is replaced by a family

$$\phi_\varepsilon(x) = \phi(x) + \varepsilon\eta(x),$$

and the field variation is

$$\delta\phi(x) = \left.\frac{\partial\phi_\varepsilon(x)}{\partial\varepsilon}\right|_{\varepsilon=0} = \eta(x).$$

Derivatives of the field vary according to

$$\delta(\partial_\mu\phi) = \partial_\mu(\delta\phi).$$

Thus the particle calculation is not an unrelated preliminary exercise. It is the one-dimensional version of the field-theoretic variation that will be used throughout the course.

What has been established

A variation is defined by embedding a reference trajectory in a one-parameter family

$$q_\varepsilon(t) = q(t) + \varepsilon\eta(t).$$

The deformation function is

$$\delta q(t) = \eta(t),$$

and its velocity variation is

$$\delta\dot{q}(t) = \dot{\eta}(t).$$

The first variation of the action is the derivative of $S[q_\varepsilon]$ with respect to the path parameter at $\varepsilon = 0$. It is linear in the deformation direction and, before integration by parts, contains both η and $\dot{\eta}$.

The next section examines the endpoint and boundary conditions that determine which variations are admissible and what happens to the boundary term in the variational calculation.

2.3 Boundary Conditions in Variational Problems

A variational problem is not defined by the action functional alone. One must also specify which trajectories are allowed to compete. The boundary data determine the admissible class of paths and therefore determine which variations may be used.

This point is essential because the derivation of the equations of motion produces a boundary term. Whether that term vanishes, survives, or imposes an additional condition depends entirely on the boundary assumptions.

What is held fixed?

Consider the action

$$S[q] = \int_{t_1}^{t_2} L(q, \dot{q}, t)\, \mathrm{d}t.$$

Before varying it, we must decide which quantities are fixed. The most common mechanical problem fixes

$$t_1, \qquad t_2, \qquad q(t_1), \qquad q(t_2).$$

Thus every admissible trajectory begins at the same initial event and ends at the same final event.

If

$$q_\varepsilon(t) = q(t) + \varepsilon\eta(t),$$

then fixed endpoint positions require

$$q_\varepsilon(t_1) = q(t_1), \qquad q_\varepsilon(t_2) = q(t_2).$$

Therefore

$$\boxed{\eta(t_1) = 0, \qquad \eta(t_2) = 0}.$$

Equivalently,

$$\delta q(t_1) = 0, \qquad \delta q(t_2) = 0.$$

These conditions apply only at the endpoints. Inside the interval, the deformation remains arbitrary.

The admissible class of trajectories

The action is evaluated on a collection of functions rather than on all imaginable curves. A typical admissible class consists of trajectories that

- are continuous on $[t_1, t_2]$;
- possess the derivatives required by the Lagrangian;
- satisfy the prescribed endpoint data;
- obey any additional constraints of the mechanical system.

For a Lagrangian depending on q and $\dot{q}$, one usually assumes at least that the trajectory is piecewise continuously differentiable. To derive the classical Euler–Lagrange equation pointwise, greater smoothness is normally assumed.

The exact function space matters in rigorous treatments. In an introductory derivation, the main requirement is that all differentiations and integrations by parts are legitimate.

Admissible and inadmissible deformations

Suppose the interval is

$$0 \le t \le T$$

and the endpoints are fixed. The function

$$\eta_1(t) = A \sin\left(\frac{\pi t}{T}\right)$$

is admissible because

$$\eta_1(0) = 0, \qquad \eta_1(T) = 0.$$

Similarly,

$$\eta_2(t) = At(T - t)$$

is admissible.

By contrast,

$$\eta_3(t) = A$$

is not admissible when the endpoint positions are fixed, because it shifts both endpoints.

The function

$$\eta_4(t) = At$$

also fails, since

$$\eta_4(T) = AT \neq 0$$

unless $A = 0$.

The stationarity condition must hold for every admissible deformation, not for deformations that violate the stated boundary data.

Fixed endpoint values do not fix endpoint slopes

From

$$\eta(t_1) = 0, \qquad \eta(t_2) = 0,$$

it does not follow that

$$\dot{\eta}(t_1) = 0, \qquad \dot{\eta}(t_2) = 0.$$

For example,

$$\eta(t) = A \sin\left(\frac{\pi(t - t_1)}{t_2 - t_1}\right)$$

vanishes at both endpoints, but its derivative is generally nonzero there.

This distinction is important. A first-order Lagrangian $L(q, \dot{q}, t)$ produces a boundary term proportional to δq, so fixing q at the endpoints is sufficient. If the Lagrangian depended on higher derivatives, additional endpoint data could be required.

How the boundary term appears

The first variation of the action is

$$\delta S = \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q}\eta + \frac{\partial L}{\partial \dot{q}}\dot{\eta} \right) \mathrm{d}t.$$

The second term contains $\dot{\eta}$. Integration by parts gives

$$\int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{q}}\dot{\eta}\,\mathrm{d}t = \left[\frac{\partial L}{\partial \dot{q}}\eta \right]_{t_1}^{t_2} - \int_{t_1}^{t_2} \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}} \right) \eta\,\mathrm{d}t.$$

Therefore

$$\boxed{\delta S = \left[\frac{\partial L}{\partial \dot{q}}\eta \right]_{t_1}^{t_2} + \int_{t_1}^{t_2} \left[\frac{\partial L}{\partial q} - \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}} \right) \right] \eta\,\mathrm{d}t}.$$

The first term is the boundary contribution. It is evaluated only at t_1 and t_2.

Why the boundary term vanishes for fixed endpoints

When the endpoint positions are fixed,

$$\eta(t_1) = 0, \qquad \eta(t_2) = 0.$$

Hence

$$\left[\frac{\partial L}{\partial \dot{q}}\eta \right]_{t_1}^{t_2} = 0.$$

The first variation then reduces to an integral over the interior:

$$\delta S = \int_{t_1}^{t_2} \left[\frac{\partial L}{\partial q} - \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}} \right) \right] \eta\,\mathrm{d}t.$$

The boundary term has not disappeared by algebraic coincidence. It vanishes because the class of admissible paths was chosen to have fixed endpoint positions.

A correct derivation should always state this reason explicitly.

What happens if an endpoint is free?

Suppose the final position is not fixed. Then $\eta(t_2)$ need not vanish. If the initial position remains fixed, we still have

$$\eta(t_1) = 0,$$

but the boundary contribution becomes

$$\left[\frac{\partial L}{\partial \dot{q}}\eta \right]_{t_1}^{t_2} = \left. \frac{\partial L}{\partial \dot{q}} \right|_{t_2} \eta(t_2).$$

For the action to be stationary for arbitrary endpoint displacement $\eta(t_2)$, one must require

$$\boxed{\left. \frac{\partial L}{\partial \dot{q}} \right|_{t_2} = 0}.$$

This is an example of a *natural boundary condition.* It is not imposed before the variation. It follows from stationarity when the corresponding endpoint value is free.

If both endpoint positions are free, analogous conditions arise at both ends.

Canonical momentum at the boundary

The quantity

$$p := \frac{\partial L}{\partial \dot{q}}$$

is the canonical momentum conjugate to q. The boundary contribution may therefore be written as

$$[p\,\delta q]_{t_1}^{t_2}.$$

This form reveals the pairing between endpoint position variations and canonical momentum. For the standard Lagrangian

$$L = \frac{m}{2}\dot{q}^2 - V(q),$$

we have

$$p = m\dot{q}.$$

Thus a free endpoint can produce a condition on the endpoint momentum.

This structure becomes even more important in field theory, where boundary terms pair field variations with normal derivatives or canonical field momenta on the boundary of a spacetime region.

Fixed times and variable times

The standard derivation assumes that t_1 and t_2 are fixed. If an endpoint time is also allowed to vary, the variation of the action contains additional terms because both the path value and the integration limit change.

Such problems lead to transversality conditions and require a more careful treatment. They are not needed for the basic derivation developed here, but they illustrate a general rule:

> Every quantity allowed to vary can contribute its own boundary term and its own boundary condition.

In the remainder of this chapter, unless stated otherwise, the endpoint times and endpoint positions are fixed.

Boundary conditions and uniqueness

The variational principle produces differential equations in the interior, but differential equations alone do not usually determine a unique trajectory. Boundary conditions or initial conditions are also required.

For a second-order equation of motion, two independent pieces of data are normally needed. One may specify

$$q(t_1) \quad \text{and} \quad q(t_2),$$

as in a fixed-endpoint variational problem, or one may specify

$$q(t_1) \quad \text{and} \quad \dot{q}(t_1),$$

as in an initial-value problem.

These are different ways of selecting a particular solution of the same differential equation. The variational derivation commonly uses endpoint data because they make the boundary term transparent.

Boundary data in field theory

For a field $\phi(x)$ defined on a spacetime region Ω, the particle endpoints are replaced by the boundary $\partial\Omega$. A standard field variation satisfies

$$\delta\phi = 0 \qquad \text{on } \partial\Omega.$$

Spacetime integration by parts then produces a surface term over $\partial\Omega$. The condition $\delta\phi = 0$ makes that term vanish, just as $\delta q(t_1) = \delta q(t_2) = 0$ removes the endpoint term in mechanics.

Other field boundary conditions are possible. If the field value is not fixed, stationarity may produce a natural boundary condition involving derivatives of the field. The particle problem is therefore the simplest model of the boundary analysis required later.

Diagnostic questions

Before discarding a boundary term, one should ask:

1. What are the boundaries of the integration region?
2. Which endpoint or boundary values are fixed?
3. Which variations are therefore required to vanish?
4. Are the endpoint times fixed or variable?
5. Does the action contain higher derivatives that require additional boundary data?
6. If a boundary value is free, what natural boundary condition follows?

These questions prevent one of the most common errors in variational calculations: setting a boundary term to zero without stating why.

What has been established

Boundary conditions are part of the definition of a variational problem. For fixed endpoint positions, admissible variations satisfy

$$\delta q(t_1) = 0, \qquad \delta q(t_2) = 0.$$

These conditions eliminate the endpoint contribution

$$\left[\frac{\partial L}{\partial \dot{q}}\delta q\right]_{t_1}^{t_2}.$$

If an endpoint is free, the boundary term does not vanish automatically and stationarity produces a natural boundary condition. The next section uses the fixed-endpoint problem to derive the Euler–Lagrange equation in full.

2.4 Euler–Lagrange Equations in Mechanics

We now derive the differential equation satisfied by a stationary trajectory. The derivation begins with the action, varies the complete path, isolates the boundary contribution, and then uses the arbitrariness of the interior deformation.

The result is the Euler–Lagrange equation. It is the basic equation of motion in Lagrangian mechanics and the direct prototype of the field equations derived later.

The variational problem

Consider one generalized coordinate $q(t)$ with action

$$S[q] = \int_{t_1}^{t_2} L(q, \dot{q}, t)\,\mathrm{d}t.$$

We compare the reference trajectory with the family

$$q_\varepsilon(t) = q(t) + \varepsilon\eta(t),$$

where the endpoint positions are fixed:

$$\eta(t_1) = 0, \qquad \eta(t_2) = 0.$$

The stationary-action condition is

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon} S[q_\varepsilon]\bigg|_{\varepsilon=0} = 0$$

for every admissible deformation $\eta(t)$.

Differentiating the action

The varied action is

$$S[q_\varepsilon] = \int_{t_1}^{t_2} L(q_\varepsilon, \dot{q}_\varepsilon, t)\,\mathrm{d}t.$$

Differentiating with respect to ε gives

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon} S[q_\varepsilon] = \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q}\frac{\partial q_\varepsilon}{\partial \varepsilon} + \frac{\partial L}{\partial \dot{q}}\frac{\partial \dot{q}_\varepsilon}{\partial \varepsilon} \right) \mathrm{d}t.$$

At $\varepsilon = 0$,

$$\frac{\partial q_\varepsilon}{\partial \varepsilon}\bigg|_{\varepsilon=0} = \eta,$$

and

$$\frac{\partial \dot{q}_\varepsilon}{\partial \varepsilon}\bigg|_{\varepsilon=0} = \dot{\eta}.$$

Therefore the first variation is

$$\delta S = \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q}\eta + \frac{\partial L}{\partial \dot{q}}\dot{\eta} \right) \mathrm{d}t.$$

All derivatives of L are evaluated along the reference trajectory $q(t)$.

Integration by parts

The term containing $\dot{\eta}$ must be rewritten so that the arbitrary function appears without a derivative. Integration by parts gives

$$\int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{q}}\dot{\eta}\,\mathrm{d}t = \left[\frac{\partial L}{\partial \dot{q}}\eta\right]_{t_1}^{t_2} - \int_{t_1}^{t_2} \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\right)\eta\,\mathrm{d}t.$$

Substituting this result into δS,

$$\delta S = \left[\frac{\partial L}{\partial \dot{q}}\eta\right]_{t_1}^{t_2} + \int_{t_1}^{t_2} \left[\frac{\partial L}{\partial q} - \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\right)\right]\eta\,\mathrm{d}t.$$

Because the endpoint positions are fixed,

$$\eta(t_1) = \eta(t_2) = 0,$$

so the boundary term vanishes. Hence

$$\delta S = \int_{t_1}^{t_2} \left[\frac{\partial L}{\partial q} - \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\partial L}{\partial \dot{q}} \right) \right] \eta(t) \, \mathrm{d}t.$$

The fundamental lemma

We now use a basic result from the calculus of variations.

If a continuous function $f(t)$ satisfies

$$\int_{t_1}^{t_2} f(t)\eta(t) \, \mathrm{d}t = 0$$

for every sufficiently smooth function $\eta(t)$ that vanishes at the endpoints, then

$$f(t) = 0$$

throughout the interior of the interval.

The reason is that η can be chosen to be localized in any small region. If f were positive or negative in some neighbourhood, one could choose η with the same sign there and obtain a nonzero integral.

Applying the lemma to the first variation yields

$$\frac{\partial L}{\partial q} - \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\partial L}{\partial \dot{q}} \right) = 0.$$

Equivalently,

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = 0}.$$

This is the Euler–Lagrange equation for one generalized coordinate.

Canonical momentum and generalized force

Define the canonical momentum conjugate to q by

$$p := \frac{\partial L}{\partial \dot{q}}.$$

Then the Euler–Lagrange equation becomes

$$\dot{p} = \frac{\partial L}{\partial q}.$$

The quantity $\partial L / \partial q$ plays the role of a generalized force. For standard Cartesian coordinates with

$$L = \frac{m}{2} \dot{q}^2 - V(q),$$

we have

$$p = m\dot{q}$$

and

$$\frac{\partial L}{\partial q} = -\frac{\mathrm{d}V}{\mathrm{d}q}.$$

Thus

$$m\ddot{q} = -\frac{\mathrm{d}V}{\mathrm{d}q},$$

which is Newton's second law with force

$$F(q) = -\frac{\mathrm{d}V}{\mathrm{d}q}.$$

The variational principle therefore reproduces the familiar mechanical equation.

Several generalized coordinates

For a system with coordinates

$$q^1(t), \dots, q^N(t),$$

the action is

$$S[\mathbf{q}] = \int_{t_1}^{t_2} L(q^1, \dots, q^N, \dot{q}^1, \dots, \dot{q}^N, t)\,\mathrm{d}t.$$

We vary each coordinate independently:

$$q^i_\varepsilon(t) = q^i(t) + \varepsilon\eta^i(t).$$

The first variation is

$$\delta S = \int_{t_1}^{t_2} \sum_{i=1}^{N} \left(\frac{\partial L}{\partial q^i}\eta^i + \frac{\partial L}{\partial \dot{q}^i}\dot{\eta}^i \right) \mathrm{d}t.$$

After integration by parts and use of the endpoint conditions,

$$\delta S = \int_{t_1}^{t_2} \sum_{i=1}^{N} \left[\frac{\partial L}{\partial q^i} - \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}^i}\right) \right] \eta^i\,\mathrm{d}t.$$

Since the functions η^i are independent, each coefficient must vanish:

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}^i}\right) - \frac{\partial L}{\partial q^i} = 0, \qquad i = 1, \dots, N}.$$

Thus one generalized coordinate produces one Euler–Lagrange equation.

Example: particle in one dimension

Let

$$L(q, \dot{q}) = \frac{m}{2}\dot{q}^2 - V(q).$$

We calculate

$$\frac{\partial L}{\partial \dot{q}} = m\dot{q},$$

so

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\right) = m\ddot{q}.$$

Also,

$$\frac{\partial L}{\partial q} = -\frac{\mathrm{d}V}{\mathrm{d}q}.$$

The Euler–Lagrange equation is therefore

$$m\ddot{q} + \frac{\mathrm{d}V}{\mathrm{d}q} = 0.$$

For the harmonic potential

$$V(q) = \frac{k}{2}q^2,$$

we obtain

$$m\ddot{q} + kq = 0.$$

For the linear potential

$$V(q) = mgq,$$

we obtain

$$m\ddot{q} + mg = 0,$$

or

$$\ddot{q} = -g.$$

Cyclic coordinates

If the Lagrangian does not depend explicitly on a coordinate q^j, then

$$\frac{\partial L}{\partial q^j} = 0.$$

The corresponding Euler–Lagrange equation becomes

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}^j}\right) = 0.$$

Therefore the conjugate momentum is conserved:

$$\boxed{p_j = \frac{\partial L}{\partial \dot{q}^j} = \text{constant}}.$$

Such a coordinate is called cyclic or ignorable. This is an early example of the relation between absence of coordinate dependence and a conservation law. The general connection between continuous symmetries and conserved quantities will later be developed through Noether's theorem.

Explicit time dependence

The Euler–Lagrange equation remains valid when the Lagrangian depends explicitly on time:

$$L = L(q, \dot{q}, t).$$

The partial derivatives $\partial L/\partial q$ and $\partial L/\partial \dot{q}$ are taken while holding the other arguments fixed. The total time derivative

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\right)$$

then includes all time dependence along the trajectory.

If L has no explicit time dependence, an energy-like quantity is conserved. That result will be established later using symmetry arguments.

A practical derivation algorithm

Given a mechanical Lagrangian, the equations of motion may be obtained as follows:

1. identify the generalized coordinates q^i;
2. write the Lagrangian $L(q^i, \dot{q}^i, t)$;
3. compute $\partial L/\partial q^i$;
4. compute $\partial L/\partial \dot{q}^i$;
5. take the total time derivative of $\partial L/\partial \dot{q}^i$;
6. substitute into the Euler–Lagrange equation;
7. simplify and check dimensions, signs, and limiting cases.

One should not replace the total derivative in step 5 by a partial derivative. The quantity $\partial L/\partial \dot{q}^i$ is evaluated along the trajectory and may depend on coordinates, velocities, and time.

Verification of a proposed solution

After deriving an equation of motion, a proposed trajectory must be checked by substitution.

For the free particle,

$$L = \frac{m}{2}\dot{q}^2,$$

and the Euler–Lagrange equation gives

$$\ddot{q} = 0.$$

The general solution is

$$q(t) = A + Bt.$$

Indeed,

$$\dot{q} = B, \qquad \ddot{q} = 0.$$

If endpoint values are specified, the constants A and B are determined by those data.

This final check is part of the calculation. Deriving the differential equation and solving it are distinct steps.

Common errors

The most frequent mistakes in the derivation are:

1. omitting the variation of $\dot{q}$;
2. writing $\delta\dot{q} = \delta q$ instead of $\delta\dot{q} = \mathrm{d}(\delta q)/\mathrm{d}t$;
3. discarding the boundary term without stating the endpoint conditions;
4. treating $\eta(t)$ as one special function rather than an arbitrary admissible deformation;
5. confusing the total derivative $\mathrm{d}/\mathrm{d}t$ with a partial derivative;
6. reversing the sign of $\partial L/\partial q$.

A reliable derivation should make each of these operations visible.

Preparation for field equations

The particle equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = 0$$

becomes, in field theory,

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0.$$

The time derivative is replaced by a spacetime divergence, the trajectory is replaced by a field, and the Lagrangian is replaced by a Lagrangian density. The logical structure of the derivation remains unchanged.

What has been established

Stationarity of the action under all smooth fixed-endpoint variations implies the Euler–Lagrange equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = 0.$$

The derivation depends on four explicit ingredients: the varied path, the commutation of variation with the time derivative, integration by parts, and the vanishing of the endpoint variation. For several independent coordinates, one equation is obtained for each coordinate.

The next section explains why generalized coordinates are useful and how constraints can be incorporated into the choice of variables.

2.5 Generalized Coordinates and Constraints

The coordinates used to describe a mechanical system need not be Cartesian positions. They should be chosen to represent the independent degrees of freedom as directly as possible. Coordinates adapted to the geometry and constraints of a system often make the Lagrangian formulation much simpler than a force-based Cartesian description.

A generalized coordinate may be a distance, an angle, or any parameter that uniquely labels the allowed configurations within the region under consideration.

Degrees of freedom

The number of degrees of freedom is the number of independent parameters required to specify a configuration.

A free particle in three-dimensional space has three degrees of freedom, which may be represented by

$$x, \qquad y, \qquad z.$$

Two free particles have six Cartesian coordinates. If constraints relate some of these coordinates, the number of independent degrees of freedom is reduced.

For example, a particle constrained to move on a sphere of radius R satisfies

$$x^2 + y^2 + z^2 = R^2.$$

Although three Cartesian coordinates appear, they are connected by one independent constraint. The particle therefore has two degrees of freedom.

A convenient choice is

$$q^1 = \theta, \qquad q^2 = \varphi,$$

with

$$\begin{aligned} x &= R \sin\theta \cos\varphi, \\ y &= R \sin\theta \sin\varphi, \\ z &= R \cos\theta. \end{aligned}$$

The constraint is then built directly into the coordinate representation.

Generalized coordinates

A set of generalized coordinates is denoted by

$$q^1, q^2, \ldots, q^N.$$

The Cartesian positions of all particles are expressed as functions of these coordinates and possibly time:

$$\mathbf{r}_a = \mathbf{r}_a(q^1, \ldots, q^N, t).$$

Here the index a labels particles. The explicit time dependence allows for moving constraints, such as a bead constrained to a wire whose shape changes with time.

The generalized coordinates need not have the dimensions of length. An angular coordinate is dimensionless in SI units, while other coordinates may have dimensions adapted to the problem.

The Euler–Lagrange equations retain the same form for every generalized coordinate:

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}^i}\right) - \frac{\partial L}{\partial q^i} = 0}.$$

This coordinate flexibility is one of the principal advantages of Lagrangian mechanics.

Velocities in generalized coordinates

If

$$\mathbf{r}_a = \mathbf{r}_a(q^1, \ldots, q^N, t),$$

then the particle velocity is obtained by the chain rule:

$$\boxed{\dot{\mathbf{r}}_a = \sum_{i=1}^{N} \frac{\partial \mathbf{r}_a}{\partial q^i}\dot{q}^i + \frac{\partial \mathbf{r}_a}{\partial t}}.$$

For a time-independent coordinate transformation,

$$\frac{\partial \mathbf{r}_a}{\partial t} = 0,$$

so

$$\dot{\mathbf{r}}_a = \sum_{i=1}^{N} \frac{\partial \mathbf{r}_a}{\partial q^i}\dot{q}^i.$$

The kinetic energy can then be expressed in terms of the generalized coordinates and velocities.

For particles with masses m_a,

$$T = \frac{1}{2}\sum_a m_a \dot{\mathbf{r}}_a^2.$$

After substitution, the kinetic energy commonly takes the form

$$T = \frac{1}{2}\sum_{i,j} g_{ij}(q)\dot{q}^i\dot{q}^j,$$

where the coefficients $g_{ij}(q)$ depend on the chosen coordinates. In geometric language, these coefficients form the metric on configuration space.

Example: a particle constrained to a circle

Consider a particle of mass m constrained to a circle of radius R in the plane. Cartesian coordinates satisfy

$$x^2 + y^2 = R^2.$$

Using Cartesian coordinates would require carrying this constraint throughout the calculation. Instead, introduce the angular coordinate θ:

$$x = R\cos\theta, \qquad y = R\sin\theta.$$

The velocity components are

$$\dot{x} = -R\sin\theta\,\dot{\theta}, \qquad \dot{y} = R\cos\theta\,\dot{\theta}.$$

Therefore

$$\begin{aligned}\dot{x}^2 + \dot{y}^2 &= R^2\left(\sin^2\theta + \cos^2\theta\right)\dot{\theta}^2\\ &= R^2\dot{\theta}^2.\end{aligned}$$

The kinetic energy is

$$T = \frac{mR^2}{2}\dot{\theta}^2.$$

If there is no potential, the Lagrangian is

$$L = \frac{mR^2}{2}\dot{\theta}^2.$$

The Euler–Lagrange equation gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(mR^2\dot{\theta}\right) = 0,$$

so

$$\ddot{\theta} = 0.$$

Uniform angular motion follows without introducing the radial constraint force. The normal force that keeps the particle on the circle does not appear explicitly because the coordinate θ already describes only allowed motion.

Holonomic constraints

A holonomic constraint can be written as a relation among coordinates and time:

$$f_\alpha(q^1, \dots, q^M, t) = 0, \qquad \alpha = 1, \dots, r.$$

If the r constraints are independent, they reduce the number of degrees of freedom from M to

$$N = M - r.$$

Examples include:

- a particle on a circle, $x^2 + y^2 - R^2 = 0$;
- a particle on a sphere, $x^2 + y^2 + z^2 - R^2 = 0$;
- two particles connected by a rigid rod, $\|\mathbf{r}_1 - \mathbf{r}_2\| - \ell = 0$.

When possible, the most direct method is to solve the constraints by introducing independent generalized coordinates.

Example: the simple pendulum

A point mass m attached to a massless rigid rod of length ℓ moves in a vertical plane. Its Cartesian coordinates may be written as

$$x = \ell \sin\theta, \qquad y = -\ell\cos\theta.$$

The rigid-length constraint

$$x^2 + y^2 = \ell^2$$

is automatically satisfied.
The velocity components are

$$\dot{x} = \ell\cos\theta\,\dot{\theta}, \qquad \dot{y} = \ell\sin\theta\,\dot{\theta}.$$

Thus

$$T = \frac{m}{2}\left(\dot{x}^2 + \dot{y}^2\right) = \frac{m\ell^2}{2}\dot{\theta}^2.$$

Taking the zero of potential energy at the lowest point,

$$V = mg\ell(1 - \cos\theta).$$

The Lagrangian is

$$L = \frac{m\ell^2}{2}\dot{\theta}^2 - mg\ell(1 - \cos\theta).$$

We calculate

$$\frac{\partial L}{\partial \dot{\theta}} = m\ell^2\dot{\theta},$$

and

$$\frac{\partial L}{\partial \theta} = -mg\ell \sin\theta.$$

The Euler–Lagrange equation gives

$$m\ell^2\ddot{\theta} + mg\ell \sin\theta = 0.$$

Dividing by $m\ell^2$,

$$\boxed{\ddot{\theta} + \frac{g}{\ell}\sin\theta = 0}.$$

The tension in the rod does not appear in this equation. It enforces the constraint but performs no virtual work along an allowed angular displacement.

Constraint forces and virtual displacements

A virtual displacement is an infinitesimal change of configuration consistent with the constraints at a fixed time. In generalized coordinates,

$$\delta \mathbf{r}_a = \sum_i \frac{\partial \mathbf{r}_a}{\partial q^i}\delta q^i.$$

For ideal holonomic constraints, the constraint forces do no virtual work:

$$\sum_a \mathbf{F}_a^{\text{constraint}} \cdot \delta \mathbf{r}_a = 0.$$

This is why such forces can disappear from the equations when independent generalized coordinates are used. The coordinates describe motion tangent to the constraint surface, while the constraint forces act in directions that maintain the restriction.

This statement must be applied with care. Friction and other non-ideal forces may perform work along allowed displacements and cannot be removed merely by changing coordinates.

Constraints imposed with Lagrange multipliers

Sometimes solving the constraints explicitly is inconvenient. One may then retain redundant coordinates and introduce Lagrange multipliers.

Suppose the coordinates q^a satisfy

$$f_\alpha(q,t) = 0.$$

Define the augmented Lagrangian

$$L_{\text{aug}} = L(q,\dot{q},t) + \sum_\alpha \lambda_\alpha(t) f_\alpha(q,t).$$

Variation with respect to λ_α gives

$$f_\alpha(q,t) = 0,$$

while variation with respect to q^a gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}^a}\right) - \frac{\partial L}{\partial q^a} = \sum_\alpha \lambda_\alpha \frac{\partial f_\alpha}{\partial q^a}.$$

The multiplier terms represent generalized constraint forces.

Example: circle constraint with a multiplier

For a free particle in the plane constrained by

$$f(x,y) = x^2 + y^2 - R^2 = 0,$$

take

$$L_{\text{aug}} = \frac{m}{2}\left(\dot{x}^2 + \dot{y}^2\right) + \lambda(t)\left(x^2 + y^2 - R^2\right).$$

The equations for x and y are

$$m\ddot{x} - 2\lambda x = 0,$$

and

$$m\ddot{y} - 2\lambda y = 0.$$

Variation with respect to λ restores the circle constraint.

The vector form is

$$m\ddot{\mathbf{r}} = 2\lambda\mathbf{r}.$$

The right-hand side is radial, as expected for the force that keeps the particle on the circle. The multiplier determines the magnitude of that constraint force.

Time-dependent constraints

A holonomic constraint may depend explicitly on time:

$$f(q,t) = 0.$$

An example is a bead constrained to a moving wire. Coordinates adapted to such a constraint can produce a Lagrangian with explicit time dependence.

In that case, energy need not be conserved because the moving constraint can transfer energy to or from the system. The Euler–Lagrange equations remain valid, but the physical interpretation of conserved quantities must account for the external time dependence.

Non-holonomic constraints

Not every constraint can be reduced to an equation involving coordinates alone. A constraint may involve velocities:

$$g_\alpha(q, \dot{q}, t) = 0.$$

Such constraints are called non-holonomic when they cannot be integrated into relations of the form $f_\alpha(q, t) = 0$.

Rolling without slipping is a standard example. The treatment of non-holonomic systems requires care because simply substituting the velocity constraint into the Lagrangian may produce incorrect equations.

The present course focuses mainly on holonomic constraints, which provide the cleanest preparation for field-theoretic variables and gauge redundancies.

Coordinate singularities and coordinate ranges

Generalized coordinates may fail to be valid globally. Spherical coordinates, for example, are singular at the poles, and an angular coordinate is periodic:

$$\varphi \sim \varphi + 2\pi.$$

A coordinate description must therefore include its range and any identifications. A singularity of the coordinate system need not represent a physical singularity of the mechanical system.

This distinction reappears in field theory, where different coordinate systems or gauge choices may describe the same physical configuration.

A practical coordinate-selection procedure

For a constrained mechanical system:

1. list the Cartesian variables;
2. identify the independent constraints;
3. count the remaining degrees of freedom;
4. choose generalized coordinates adapted to the allowed configurations;
5. express positions and velocities in those coordinates;
6. compute T and V;
7. form $L = T - V$;
8. apply one Euler–Lagrange equation to each independent coordinate.

If the constraints cannot be solved conveniently, introduce Lagrange multipliers and vary the augmented action.

Connection with field variables

A field may be regarded as a system with one or more generalized coordinates at every point of space. The field components are the dynamical variables, and relations among them may play the role of constraints or redundancies.

The lesson from mechanics is that the choice of variables matters. Variables adapted to the structure of the theory can remove unnecessary equations and expose the true independent degrees of freedom.

In gauge theories, however, the variables often contain deliberate redundancy rather than an ordinary mechanical constraint. Distinguishing physical degrees of freedom from descriptive redundancy will become a central issue later.

What has been established

Generalized coordinates parameterize the independent configurations of a mechanical system. Holonomic constraints reduce the number of degrees of freedom and can often be incorporated directly into the coordinates. When this is done, ideal constraint forces need not appear explicitly in the Euler–Lagrange equations.

Alternatively, constraints may be enforced with Lagrange multipliers, whose equations restore the constraints and whose contributions represent constraint forces.

The next section examines integration by parts and boundary terms as a general mathematical mechanism, preparing the transition from one-dimensional mechanical actions to spacetime field actions.

2.6 Integration by Parts and Boundary Terms

Integration by parts is the operation that converts the variation of a derivative into the derivative of a coefficient. In variational mechanics, this step separates the first variation into two distinct pieces:

- an interior term that produces the differential equation of motion;
- a boundary term that records what happens at the endpoints.

The same structure reappears in every later field-theoretic derivation. Understanding it in one time dimension is therefore more important than memorizing the final Euler–Lagrange formula.

The ordinary integration-by-parts identity

For sufficiently differentiable functions $f(t)$ and $g(t)$,

$$\int_{t_1}^{t_2} f(t)\dot{g}(t)\,\mathrm{d}t = [f(t)g(t)]_{t_1}^{t_2} - \int_{t_1}^{t_2} \dot{f}(t)g(t)\,\mathrm{d}t.$$

This follows from the product rule

$$\frac{\mathrm{d}}{\mathrm{d}t}(fg) = \dot{f}g + f\dot{g}.$$

Rearranging,

$$f\dot{g} = \frac{\mathrm{d}}{\mathrm{d}t}(fg) - \dot{f}g,$$

and integrating over the interval gives the identity.

The notation

$$[fg]_{t_1}^{t_2} = f(t_2)g(t_2) - f(t_1)g(t_1)$$

makes the endpoint contribution explicit.

Application to the first variation

For the action

$$S[q] = \int_{t_1}^{t_2} L(q, \dot{q}, t)\, \mathrm{d}t,$$

the first variation is

$$\delta S = \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q} \delta q + \frac{\partial L}{\partial \dot{q}} \delta \dot{q} \right) \mathrm{d}t.$$

Since

$$\delta \dot{q} = \frac{\mathrm{d}}{\mathrm{d}t}(\delta q),$$

the second term has the form required for integration by parts. Set

$$f(t) = \frac{\partial L}{\partial \dot{q}}, \qquad g(t) = \delta q(t).$$

Then

$$\int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{q}} \delta \dot{q}\, \mathrm{d}t = \left[\frac{\partial L}{\partial \dot{q}} \delta q \right]_{t_1}^{t_2} - \int_{t_1}^{t_2} \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\partial L}{\partial \dot{q}} \right) \delta q\, \mathrm{d}t.$$

Therefore

$$\boxed{\delta S = \left[\frac{\partial L}{\partial \dot{q}} \delta q \right]_{t_1}^{t_2} + \int_{t_1}^{t_2} \left[\frac{\partial L}{\partial q} - \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\partial L}{\partial \dot{q}} \right) \right] \delta q\, \mathrm{d}t}.$$

The derivative has been transferred from δq to $\partial L / \partial \dot{q}$. This is the decisive step in the derivation.

Why the derivative must be moved

Before integration by parts, the first variation contains both δq and $\delta \dot{q}$. These are not independent functions because

$$\delta \dot{q} = \frac{\mathrm{d}}{\mathrm{d}t}(\delta q).$$

The fundamental lemma of the calculus of variations applies to an integral of the form

$$\int f(t) \delta q(t)\, \mathrm{d}t.$$

Integration by parts rewrites the complete interior contribution in precisely this form. The arbitrariness of δq can then be used to conclude that its coefficient vanishes.

Without this step, one cannot correctly infer separate conditions from the coefficients of δq and $\delta \dot{q}$.

The boundary term as a separate object

The boundary contribution is

$$B = \left[\frac{\partial L}{\partial \dot{q}} \delta q \right]_{t_1}^{t_2}.$$

Using the canonical momentum

$$p = \frac{\partial L}{\partial \dot{q}},$$

we may write

$$B = [p\,\delta q]_{t_1}^{t_2}.$$

This term contains no time integral. It depends only on endpoint data.
For fixed endpoint positions,

$$\delta q(t_1) = \delta q(t_2) = 0,$$

so

$$B = 0.$$

For free endpoints, the term survives and produces natural boundary conditions. Thus one should never write "the boundary term is zero" without first stating the conditions that make it zero.

A direct example

Consider

$$L = \frac{m}{2}\dot{q}^2 - V(q).$$

The first variation is

$$\delta S = \int_{t_1}^{t_2} \left(-V'(q)\delta q + m\dot{q}\,\delta\dot{q}\right)\,\mathrm{d}t.$$

Integrating the second term by parts,

$$\int_{t_1}^{t_2} m\dot{q}\,\delta\dot{q}\,\mathrm{d}t = [m\dot{q}\,\delta q]_{t_1}^{t_2} - \int_{t_1}^{t_2} m\ddot{q}\,\delta q\,\mathrm{d}t.$$

Hence

$$\delta S = [m\dot{q}\,\delta q]_{t_1}^{t_2} - \int_{t_1}^{t_2} \left(m\ddot{q} + V'(q)\right)\delta q\,\mathrm{d}t.$$

For fixed endpoints, the first term vanishes. Stationarity then implies

$$m\ddot{q} + V'(q) = 0.$$

Every sign in this equation can be traced back to the product rule and the minus sign introduced by integration by parts.

Sign checking

A reliable way to check the sign is to start from

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\delta q\right) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\right)\delta q + \frac{\partial L}{\partial \dot{q}}\delta\dot{q}.$$

Solving for the last term,

$$\frac{\partial L}{\partial \dot{q}}\delta\dot{q} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\delta q\right) - \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\right)\delta q.$$

The minus sign is therefore unavoidable.

Adding a total time derivative to the Lagrangian

Suppose two Lagrangians differ by a total derivative:

$$L'(q,\dot q,t) = L(q,\dot q,t) + \frac{\mathrm{d}}{\mathrm{d}t}F(q,t).$$

Their actions satisfy

$$\begin{aligned} S'[q] &= \int_{t_1}^{t_2} L'\,\mathrm{d}t \\ &= S[q] + \int_{t_1}^{t_2} \frac{\mathrm{d}F}{\mathrm{d}t}\,\mathrm{d}t \\ &= S[q] + F(q(t_2),t_2) - F(q(t_1),t_1). \end{aligned}$$

Thus the two actions differ only by an endpoint term.

For fixed endpoint values and fixed endpoint times, the variation of this additional term vanishes. Therefore L and L' produce the same Euler–Lagrange equations.

This establishes an important equivalence:

$$\boxed{L \sim L + \frac{\mathrm{d}F}{\mathrm{d}t}}.$$

The Lagrangian is not unique. Different Lagrangians can encode the same equations of motion when they differ by a total derivative.

Example of equivalent Lagrangians

Consider

$$L = \frac{m}{2}\dot q^2 - V(q)$$

and choose

$$F(q,t) = \alpha q t,$$

where α is constant. Then

$$\frac{\mathrm{d}F}{\mathrm{d}t} = \alpha q + \alpha t\dot q.$$

The modified Lagrangian is

$$L' = \frac{m}{2}\dot q^2 - V(q) + \alpha q + \alpha t\dot q.$$

Although the individual partial derivatives of L' differ from those of L, the extra contributions cancel in the Euler–Lagrange equation.

Indeed,

$$\frac{\partial L'}{\partial q} = -V'(q) + \alpha,$$

and

$$\frac{\partial L'}{\partial \dot q} = m\dot q + \alpha t.$$

Therefore

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L'}{\partial \dot{q}}\right) = m\ddot{q} + \alpha.$$

The Euler–Lagrange equation becomes

$$m\ddot{q} + \alpha - (-V'(q) + \alpha) = 0,$$

so

$$m\ddot{q} + V'(q) = 0.$$

The dynamics is unchanged.

Boundary terms are not always negligible

A boundary term may be physically or mathematically important when

- endpoint values are not fixed;
- the integration region has a physical boundary;
- one studies conserved charges or canonical momenta;
- the action contains higher derivatives;
- the theory is defined on a noncompact region and fall-off conditions matter;
- surface contributions encode measurable quantities.

Thus "boundary term" does not mean "unimportant term." It means a term supported on the boundary rather than in the interior.

Repeated integration by parts

If an action depends on higher derivatives, for example

$$S[q] = \int L(q, \dot{q}, \ddot{q}, t)\,\mathrm{d}t,$$

then varying $\ddot{q}$ produces

$$\delta\ddot{q} = \frac{\mathrm{d}^2}{\mathrm{d}t^2}(\delta q).$$

Two integrations by parts are required to remove derivatives from δq. The resulting boundary terms contain both δq and $\delta\dot{q}$.

This explains why higher-derivative actions generally require more boundary data and produce higher-order equations of motion.

The present course will mainly use first-derivative Lagrangians, but the pattern is worth recognizing.

Several coordinates

For generalized coordinates q^i,

$$\delta S = \int_{t_1}^{t_2} \sum_i \left(\frac{\partial L}{\partial q^i} \delta q^i + \frac{\partial L}{\partial \dot{q}^i} \delta \dot{q}^i \right) \mathrm{d}t.$$

Integrating each velocity term by parts gives

$$\delta S = \left[\sum_i \frac{\partial L}{\partial \dot{q}^i} \delta q^i \right]_{t_1}^{t_2} + \int_{t_1}^{t_2} \sum_i \left[\frac{\partial L}{\partial q^i} - \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\partial L}{\partial \dot{q}^i} \right) \right] \delta q^i \, \mathrm{d}t.$$

The boundary term is a sum of momentum-coordinate pairings:

$$\left[\sum_i p_i \delta q^i \right]_{t_1}^{t_2}.$$

The field-theory analogue

For a scalar field with action

$$S[\phi] = \int_\Omega \mathcal{L}(\phi, \partial_\mu \phi, x) \, \mathrm{d}^4 x,$$

the first variation contains

$$\int_\Omega \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\mu (\delta \phi) \, \mathrm{d}^4 x.$$

Spacetime integration by parts gives

$$\begin{aligned} &\int_\Omega \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\mu (\delta \phi) \, \mathrm{d}^4 x \\ &\quad = \int_{\partial\Omega} \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta\phi \, n_\mu \, \mathrm{d}\Sigma - \int_\Omega \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) \delta\phi \, \mathrm{d}^4 x. \end{aligned}$$

The endpoint term of mechanics becomes a surface integral over the boundary $\partial\Omega$. The one-dimensional formula is therefore not merely analogous to the field calculation; it is its simplest special case.

A diagnostic procedure

Whenever integration by parts appears in a variational derivation:

1. identify the derivative acting on the variation;
2. write the product-rule identity explicitly;
3. separate the boundary and interior terms;
4. state the boundary conditions;
5. justify whether the boundary term vanishes;
6. check the sign of the interior derivative;
7. only then apply the fundamental lemma.

Following this sequence prevents hidden assumptions and sign errors.

What has been established

Integration by parts converts the derivative of a variation into an interior derivative acting on the canonical momentum and an explicit boundary contribution. The interior term yields the Euler–Lagrange equation, while the boundary term records the endpoint assumptions.

Lagrangians that differ by a total time derivative produce actions that differ only by endpoint terms and therefore give the same fixed-endpoint equations of motion.

The next section applies the full method to several concrete particle models, from free motion to oscillation and constrained dynamics.

2.7 Worked Particle Models

The purpose of this section is to apply the variational method as a complete calculation. For each model, we identify the coordinate, write the Lagrangian, derive the Euler–Lagrange equation, solve or simplify the equation, and interpret the result.

The examples are deliberately standard. Their value lies not in novelty but in making the procedure operational.

Model 1: the free particle

Consider a particle of mass m moving on a line with no potential energy. The generalized coordinate is $q(t)$, and the Lagrangian is

$$L(q, \dot{q}) = \frac{m}{2}\dot{q}^2.$$

The required derivatives are

$$\frac{\partial L}{\partial q} = 0,$$

and

$$\frac{\partial L}{\partial \dot{q}} = m\dot{q}.$$

Therefore

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\right) = m\ddot{q}.$$

The Euler–Lagrange equation gives

$$m\ddot{q} = 0,$$

or

$$\boxed{\ddot{q} = 0}.$$

Integrating once,

$$\dot{q} = v_0,$$

where v_0 is constant. Integrating again,

$$\boxed{q(t) = q_0 + v_0 t}.$$

The canonical momentum is

$$p = \frac{\partial L}{\partial \dot{q}} = m\dot{q},$$

and the equation of motion states

$$\dot{p} = 0.$$

Thus free motion corresponds to constant momentum and constant velocity.

Fixed-endpoint solution

Suppose

$$q(t_1) = q_1, \qquad q(t_2) = q_2.$$

The solution satisfying these conditions is

$$\boxed{q(t) = q_1 + \frac{q_2 - q_1}{t_2 - t_1}(t - t_1)}.$$

The physical path is the straight line in the (t, q)-plane connecting the two endpoint events.

Check by substitution

Differentiating twice,

$$\dot{q} = \frac{q_2 - q_1}{t_2 - t_1}, \qquad \ddot{q} = 0.$$

The trajectory therefore satisfies the Euler–Lagrange equation exactly.

Model 2: a particle in a general potential

Let

$$L(q, \dot{q}) = \frac{m}{2}\dot{q}^2 - V(q).$$

Then

$$\frac{\partial L}{\partial q} = -V'(q),$$

and

$$\frac{\partial L}{\partial \dot{q}} = m\dot{q}.$$

The Euler–Lagrange equation is

$$m\ddot{q} + V'(q) = 0.$$

Defining the force by

$$F(q) = -V'(q),$$

we obtain

$$\boxed{m\ddot{q} = F(q)}.$$

This is Newton's second law. The Lagrangian and Newtonian formulations therefore describe the same dynamics for conservative forces.

Energy check

Multiply the equation of motion by $\dot{q}$:

$$m\ddot{q}\dot{q} + V'(q)\dot{q} = 0.$$

Using

$$m\ddot{q}\dot{q} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{m}{2}\dot{q}^2\right),$$

and

$$V'(q)\dot{q} = \frac{\mathrm{d}V}{\mathrm{d}t},$$

we find

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{m}{2}\dot{q}^2 + V(q)\right) = 0.$$

Thus

$$\boxed{E = \frac{m}{2}\dot{q}^2 + V(q) = \text{constant}}.$$

This conservation law provides a useful independent check of solutions.

Model 3: motion in a uniform gravitational field

Choose the vertical coordinate y, positive upward. The potential energy is

$$V(y) = mgy.$$

The Lagrangian is

$$L(y, \dot{y}) = \frac{m}{2}\dot{y}^2 - mgy.$$

We calculate

$$\frac{\partial L}{\partial y} = -mg,$$

and

$$\frac{\partial L}{\partial \dot{y}} = m\dot{y}.$$

Therefore

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{y}}\right) = m\ddot{y}.$$

The Euler–Lagrange equation gives

$$m\ddot{y} + mg = 0.$$

Dividing by m,

$$\boxed{\ddot{y} = -g}.$$

Integrating,

$$\dot{y}(t) = v_0 - gt,$$

and

$$\boxed{y(t) = y_0 + v_0 t - \frac{g}{2}t^2}.$$

Physical interpretation

The mass cancels from the equation of motion. In this idealized model, all particles have the same gravitational acceleration regardless of mass.

The conserved energy is

$$E = \frac{m}{2}\dot{y}^2 + mgy.$$

Substituting the solution verifies that the decrease in kinetic energy during upward motion equals the increase in gravitational potential energy.

Model 4: the harmonic oscillator

Consider a particle attached to a spring with spring constant k. The potential energy is

$$V(q) = \frac{k}{2}q^2.$$

The Lagrangian is

$$L(q, \dot{q}) = \frac{m}{2}\dot{q}^2 - \frac{k}{2}q^2.$$

We have

$$\frac{\partial L}{\partial q} = -kq,$$

and

$$\frac{\partial L}{\partial \dot{q}} = m\dot{q}.$$

The Euler–Lagrange equation gives

$$m\ddot{q} + kq = 0.$$

Define

$$\omega^2 = \frac{k}{m}.$$

Then

$$\boxed{\ddot{q} + \omega^2 q = 0}.$$

The general solution is

$$\boxed{q(t) = A\cos(\omega t) + B\sin(\omega t)}.$$

Equivalently,

$$q(t) = C\cos(\omega t - \varphi).$$

Verification

Differentiating twice,

$$\ddot{q}(t) = -\omega^2 \left[A\cos(\omega t) + B\sin(\omega t)\right] = -\omega^2 q(t).$$

Thus the proposed solution satisfies the equation.

Energy

The conserved energy is

$$E = \frac{m}{2}\dot{q}^2 + \frac{k}{2}q^2.$$

For

$$q(t) = C\cos(\omega t - \varphi),$$

we have

$$\dot{q}(t) = -C\omega\sin(\omega t - \varphi).$$

Therefore

$$\begin{aligned}
E &= \frac{m}{2}C^2\omega^2\sin^2(\omega t - \varphi) + \frac{k}{2}C^2\cos^2(\omega t - \varphi) \\
&= \frac{kC^2}{2}\left[\sin^2(\omega t - \varphi) + \cos^2(\omega t - \varphi)\right] \\
&= \boxed{\frac{kC^2}{2}}.
\end{aligned}$$

The total energy remains constant while kinetic and potential energies exchange periodically.

Model 5: a particle in two-dimensional polar coordinates

Consider a particle moving in a plane. Introduce polar coordinates

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

The velocity components are

$$\dot{x} = \dot{r}\cos\theta - r\dot{\theta}\sin\theta,$$

and

$$\dot{y} = \dot{r}\sin\theta + r\dot{\theta}\cos\theta.$$

Squaring and adding,

$$\dot{x}^2 + \dot{y}^2 = \dot{r}^2 + r^2\dot{\theta}^2.$$

For a central potential $V(r)$, the Lagrangian is

$$\boxed{L(r,\theta,\dot{r},\dot{\theta}) = \frac{m}{2}\left(\dot{r}^2 + r^2\dot{\theta}^2\right) - V(r)}.$$

Angular equation

The coordinate θ does not appear explicitly in L:

$$\frac{\partial L}{\partial \theta} = 0.$$

The conjugate momentum is

$$p_\theta = \frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta}.$$

The Euler–Lagrange equation gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(mr^2\dot{\theta}\right) = 0.$$

Therefore

$$\boxed{mr^2\dot{\theta} = \ell = \text{constant}}.$$

The conserved quantity ℓ is the angular momentum about the origin.

Radial equation

For the radial coordinate,

$$\frac{\partial L}{\partial \dot{r}} = m\dot{r},$$

so

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{r}}\right) = m\ddot{r}.$$

Also,

$$\frac{\partial L}{\partial r} = mr\dot{\theta}^2 - V'(r).$$

The radial Euler–Lagrange equation is

$$m\ddot{r} - mr\dot{\theta}^2 + V'(r) = 0.$$

Equivalently,

$$\boxed{m\ddot{r} = mr\dot{\theta}^2 - V'(r)}.$$

The term

$$mr\dot{\theta}^2$$

arises from the coordinate dependence of the kinetic energy. It is not inserted as an additional force by hand.

Effective potential

Using

$$\dot{\theta} = \frac{\ell}{mr^2},$$

the energy becomes

$$E = \frac{m}{2}\dot{r}^2 + \frac{\ell^2}{2mr^2} + V(r).$$

Define the effective potential

$$\boxed{V_{\text{eff}}(r) = V(r) + \frac{\ell^2}{2mr^2}}.$$

Then the radial motion takes the one-dimensional energy form

$$E = \frac{m}{2}\dot{r}^2 + V_{\text{eff}}(r).$$

This reduction shows how a multidimensional problem can be simplified using a cyclic coordinate and its conserved momentum.

Model 6: the simple pendulum and the small-angle limit

For a pendulum of length ℓ and mass m, the Lagrangian is

$$L = \frac{m\ell^2}{2}\dot{\theta}^2 - mg\ell(1 - \cos\theta).$$

The equation of motion is

$$\boxed{\ddot{\theta} + \frac{g}{\ell}\sin\theta = 0}.$$

This equation is nonlinear because of the sine function.
For small angular displacements,

$$|\theta| \ll 1,$$

we use

$$\sin\theta \approx \theta.$$

The equation becomes

$$\ddot{\theta} + \frac{g}{\ell}\theta = 0.$$

This is the harmonic-oscillator equation with

$$\omega_0 = \sqrt{\frac{g}{\ell}}.$$

The approximate solution is

$$\boxed{\theta(t) = A\cos(\omega_0 t) + B\sin(\omega_0 t)}.$$

The corresponding small-amplitude period is

$$\boxed{T = 2\pi\sqrt{\frac{\ell}{g}}}.$$

This example illustrates linearization: a nonlinear equation is approximated near an equilibrium by retaining only the leading term in a series expansion.

Comparing the models

The examples reveal a common pattern.

1. The free particle has a cyclic position coordinate and conserved linear momentum.
2. A potential introduces a generalized force through $-V'(q)$.
3. A quadratic potential produces linear harmonic motion.
4. Curvilinear coordinates make the kinetic energy coordinate dependent.
5. A cyclic angular coordinate produces conserved angular momentum.
6. A nonlinear system may reduce to a linear oscillator near equilibrium.

These are not isolated tricks. The same structural ideas recur in field theory: kinetic terms, potentials, symmetries, conserved quantities, coordinate choices, and linearization around simple backgrounds.

A complete checking procedure

For every model derived from an action, one should check:

1. Are all terms in the Lagrangian dimensionally consistent?
2. Were the generalized coordinates and constraints identified correctly?
3. Was the total time derivative in the Euler–Lagrange equation calculated fully?
4. Does the sign of the force agree with the physical potential?
5. Does a proposed solution satisfy the differential equation?
6. Are conserved quantities constant along the solution?
7. Does the result reduce correctly in a simple limit?

These checks often reveal errors more quickly than repeating the entire derivation.

Chapter summary

This chapter developed the action principle first as a mathematical construction and then as a practical method of deriving dynamics.

The action is a functional of the complete trajectory:

$$S[q] = \int_{t_1}^{t_2} L(q, \dot{q}, t)\, dt.$$

A varied path is written as

$$q_\varepsilon(t) = q(t) + \varepsilon\eta(t),$$

and fixed endpoint positions imply

$$\eta(t_1) = \eta(t_2) = 0.$$

Integration by parts separates the variation into a boundary term and an interior term. Stationarity for arbitrary admissible deformations produces the Euler–Lagrange equation

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}^i}\right) - \frac{\partial L}{\partial q^i} = 0}.$$

Generalized coordinates incorporate constraints and adapt the description to the geometry of the system. The worked models showed how the same procedure produces free motion, uniform acceleration, harmonic oscillation, angular-momentum conservation, effective radial dynamics, and the small-angle pendulum.

The next chapter extends every part of this construction from particle trajectories to fields. The generalized coordinate $q(t)$ becomes a field $\phi(x)$, the Lagrangian becomes a Lagrangian density, the time integral becomes a spacetime integral, and ordinary integration by parts becomes spacetime integration by parts.

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